Etch

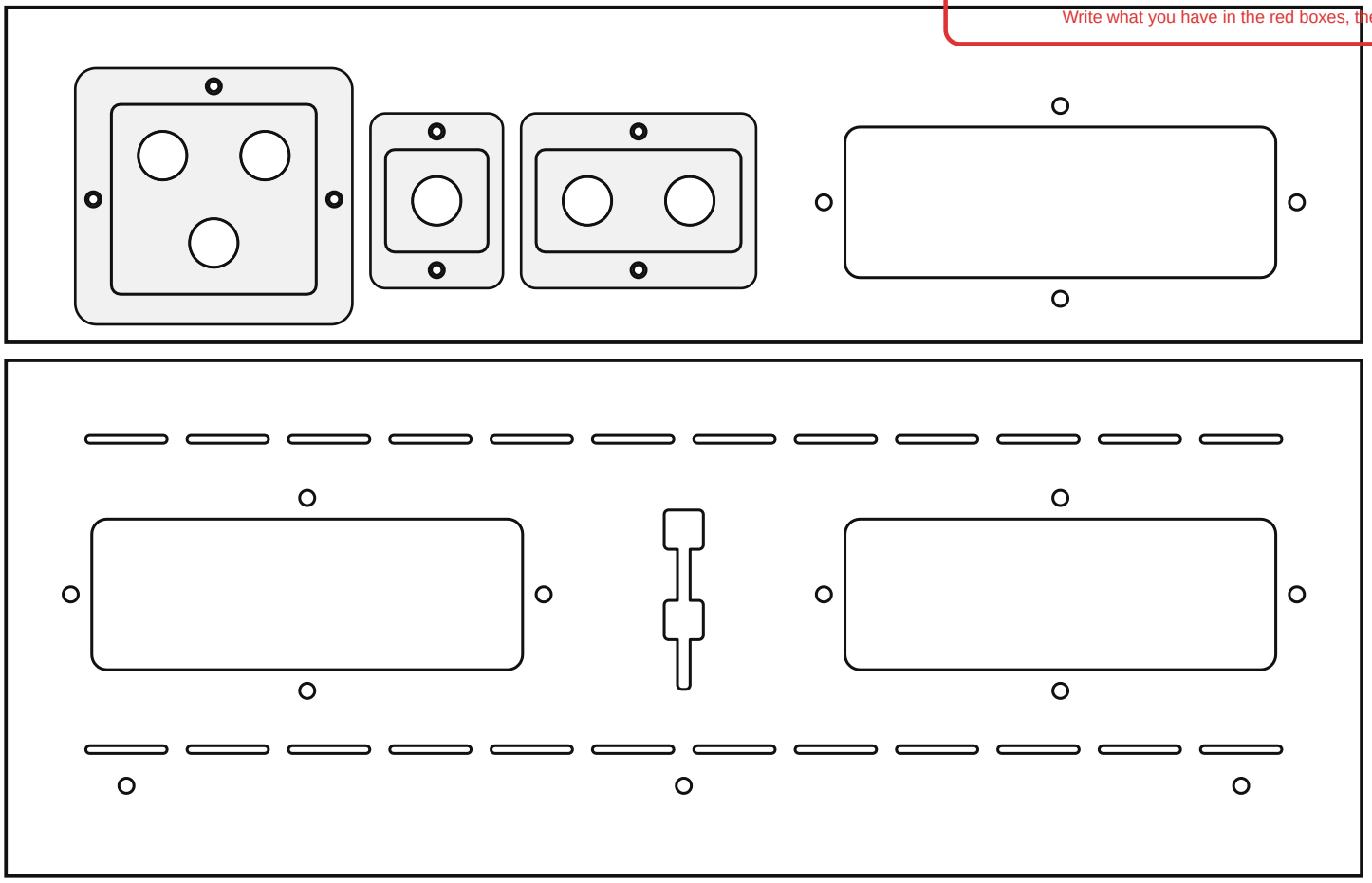
Build book · pen plotter enclosure

Power box below · control box on top
450 × 295 inside · 1.2 mm 304 stainless

One complete A3 artwork. Built once, in the case.

AUDIT COPY

Write what you have in the red boxes, then send the counts back.



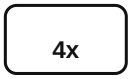
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5	**0.2a**	Rivnuts · control box · 1 parts	☐
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10	**1.3**	PSUs and buck · 6 parts	☐
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page	step	title	done
	STAGE 2 · CONTROL BOX		
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25	**2.5a**	GX sockets into the plates · 4 parts	☐
26	**2.5b**	GX plates onto the front · 1 parts	☐
27	**2.6**	Receiving panel · 4 parts	☐
28	**2.7**	USB / antenna panel · 3 parts	☐
29	**2.8a**	Power wiring · duct 1 · 3 parts	☐
30	**2.8b**	Power wiring · servo supply · 1 parts	☐
31	**2.9a**	Signals · board to drivers · 3 parts	☐
32	**2.9b**	Signals · socket pins · 3 parts	☐
33	**2.10**	Fan · control box · 4 parts	☐
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35	**3.1**	Window panels · 2 parts	☐
36	**3.2**	Lids · 2 parts	☐
37	**3.3**	Stack · 2 parts	☐
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39	**4.1**	Checks before motors · 1 parts	☐
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41	**BOM 1**		
42	**BOM 2**		
43	**BOM 3**		
44	**Prints 1**		

How to read

1. Build top to bottom, once, inside the case.
2. Before each session, read the next two pages and buy or print anything missing (BOM and Prints at the back).
3. The Etch Build Book doc is the master: if a page and the doc disagree, the doc wins and this page gets redrawn.



Parts box: what this step uses, how many, and its BOM or Prints ID (Bxx, Pxx).



Black lines: what you work on in this step. Grey lines: the rest of the face, for orientation.



Red: where to put it.



FRONT arrow: which way the box's front faces in the drawing.



a, b: sub-steps, in order.

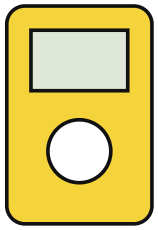


Check: do not start the next page until this passes.

⚠ unverified: a guess or an unmeasured number. Measure first.

🚫 gate: a decision is needed before this step.

0.1 Tools and mains rules



1x
multimeter,
beep + 0.01 V

B60

have ☐

rivnut tool

1x
rivnut tool, M3
+ M4

B61

have ☐

ferrule crimper

1x
ferrule crimper
0.25–6 mm²

B62

have ☐

ring crimper

1x
crimper,
insulated rings

B63

have ☐

labels

1x
wire labels

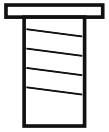
B64

have ☐

1. Lay out the tools in the parts box before starting.
2. Label every wire at both ends as you cut it.
3. Also on the bench: stripper, screwdrivers, 2.5 / 3 mm hex keys, 5.5 / 7 / 10 mm spanners, iron with a heat-set tip.

1. **Unplugged means unplugged.**
   The IEC lead is out of the wall and in sight before a hand goes into the power box.
2. **Ferrule every stranded wire under a screw.**
   Never tin a wire with solder under a screw: it creeps and loosens.
3. **Brown = live, blue = neutral, green/yellow = earth.**
   Never use green/yellow for anything else.
4. **Earth first, earth last.**
   Fitted first, removed last, and longer than L and N so it pulls tight last.
5. **DC 0 V never touches earth.**
   G1 and G2 stay off the box metal; the laptop's USB already earths 0 V.
6. **Shroud every mains terminal**
   before the first power-on.
7. **Never plug or unplug a motor with 24 V on.**
   A TB6600 whose motor is pulled live often dies.

0.2a Rivnuts · control box



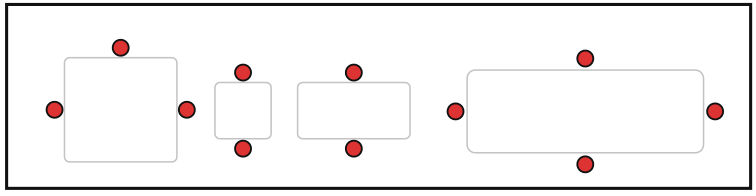
21x
M3 rivnut, Ø5 hole, flange Ø8

B29

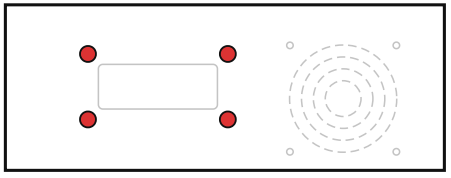
have ☐

1. Set 21 M3 rivnuts in the control box: 15 in the walls (red) and 6 in the lips for the lid.
2. Flange on the outside, bulge on the inside. Squeeze the tool until it stops; don't over-pull.
3. The 4 lid-top accessory holes stay empty for now.

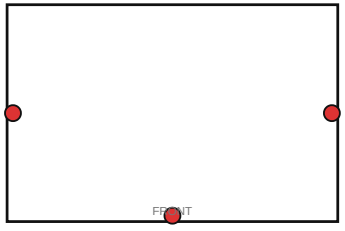
Total both boxes: 39. Buy 50.



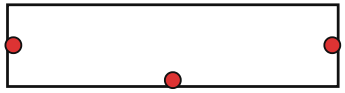
front, outside



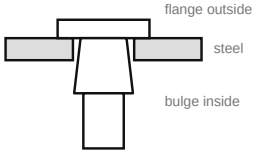
left, outside



top lips, from above



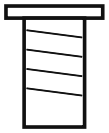
back lips, from behind



☑ **Each takes an M3 screw by hand and doesn't spin**

0.2b

Rivnuts · power box



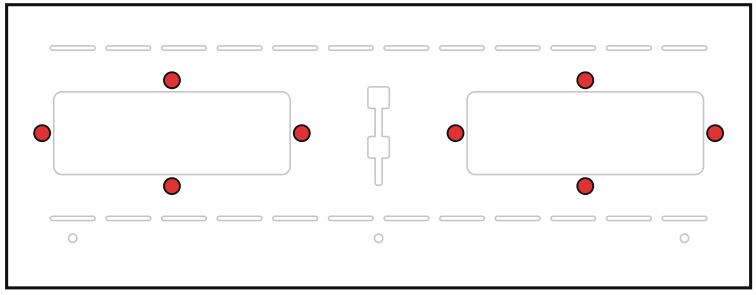
18x
M3 rivnut, Ø5 hole, flange Ø8

B29

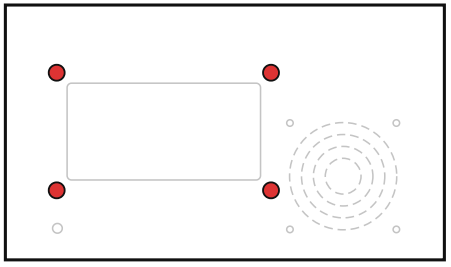
have ☐

1. Set 18 M3 rivnuts in the power box: 12 in the walls (red) and 6 in the lips for the lid.
2. Flange on the outside, bulge on the inside. Squeeze the tool until it stops; don't over-pull.
3. Leave the Ø5 accessory holes empty for now.

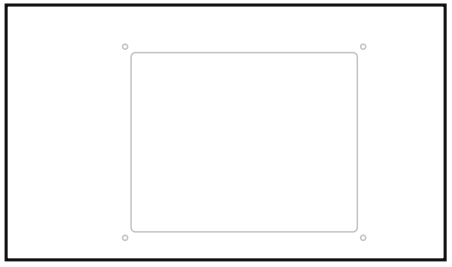
Total both boxes: 39. Buy 50.



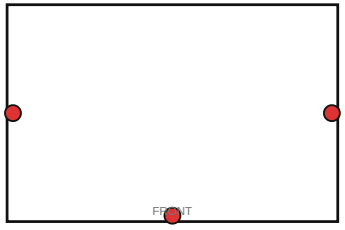
front, outside



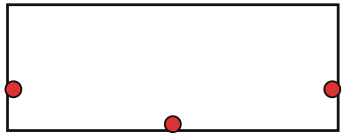
left, outside



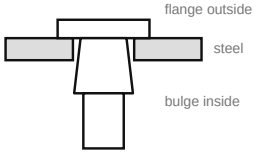
right: banana panel uses screws, no rivnuts



top lips, from above

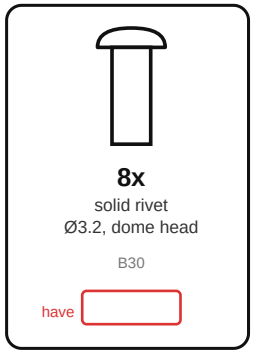


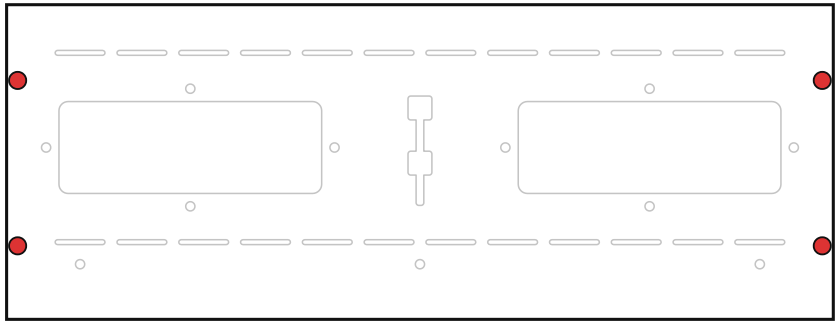
back lips, from behind



☑ **Each takes an M3 screw by hand and doesn't spin**

0.3 Corner rivets

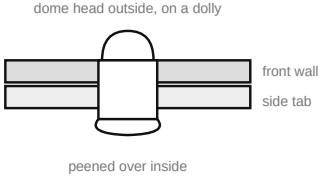




front corners, outside (both boxes)

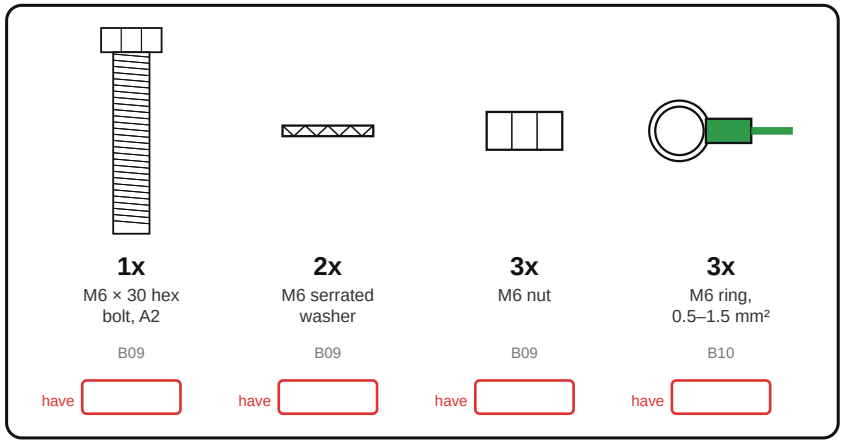
1. 8 solid Ø3.2 rivets: 2 per front corner tab, 2 tabs per box, 2 boxes.
2. Squeeze the tab against the front wall with a clamp first; the gap closes by the rivet, not the hammer.
3. Dome head outside on a dolly. Peen the tail over from inside with light blows.

Rivet positions drawn approximately: use the holes in the steel.



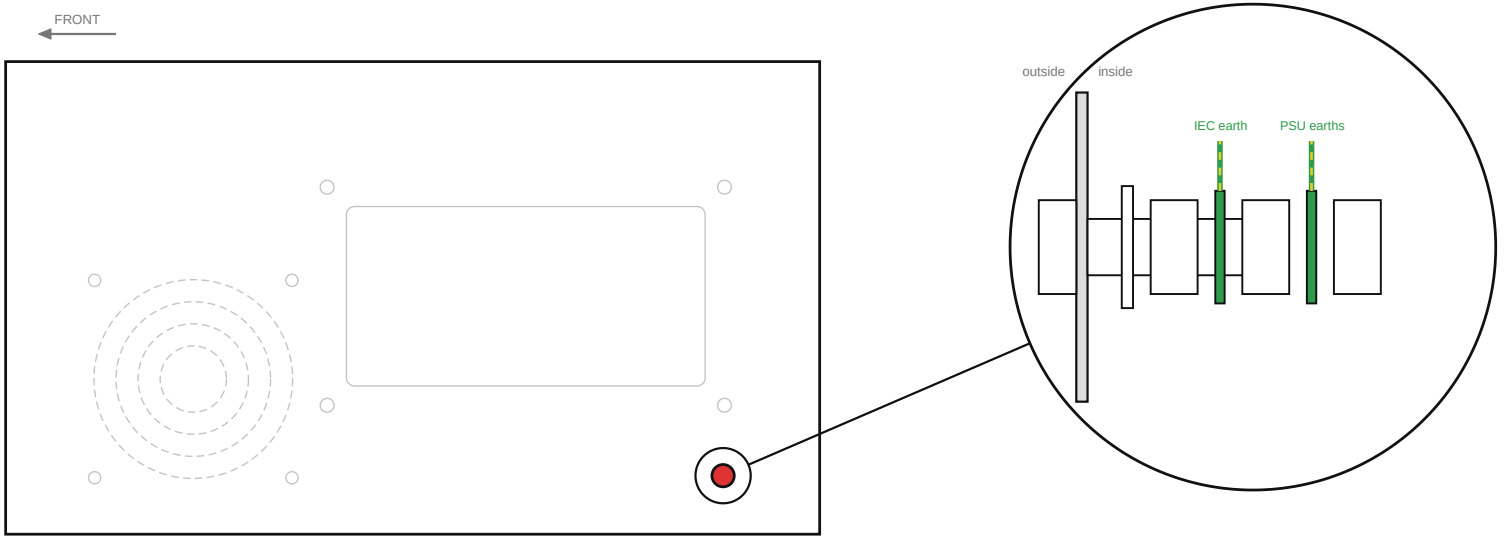
☑ **Tab tight against the wall, no rocking**

1.1 Earth stud



1. M6 bolt through the Ø6.5 hole low on the left wall, head outside.
2. Inside: serrated washer, nut. Tighten hard: this is the bond to the box.
3. The IEC earth ring goes on next with its own nut (1.5). PSU earth rings go above it under a third nut.

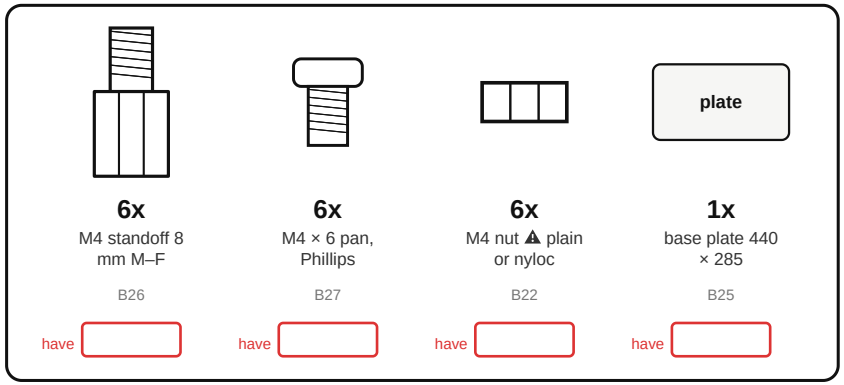
⚠ scrape any film off under the serrated washer



Power box, left wall, from inside

☑ **Beep: stud ↔ bare steel anywhere on the body**

1.2 Base plate · power box

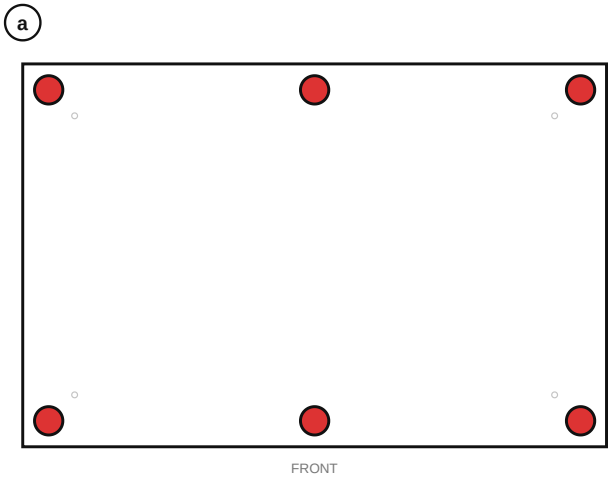


1. a Six standoffs onto the floor: M4 × 6 pan screw up from underneath into the female end.
2. b Plate on, studs through its standoff holes, a nut on each. The plate comes out from inside by undoing 6 nuts. Set the PSU-bracket M4 rivnuts first (1.3).

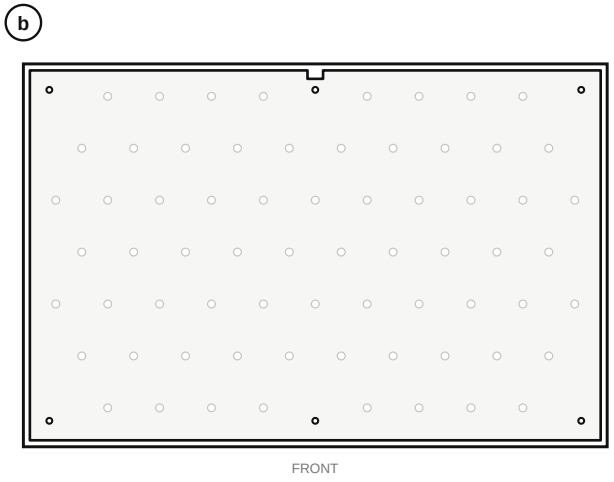
Plates: 2 mm mild steel, already cut

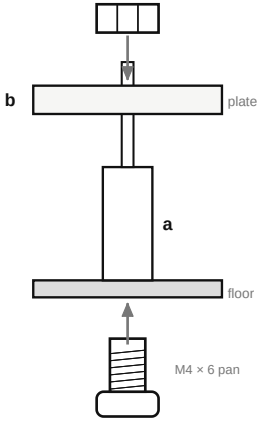
Power: PSU panel's bottom screws clear the plate by only 0.3 mm

each of 6



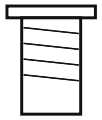
Power box floor, from above

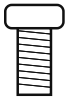




☑ **Plate doesn't rock · no screw head below the feet**

1.3 PSUs and buck

24 V PSU	12 V PSU	buck	bracket	
1x	**1x**	**1x**	**2 sets**	**8x**
LRS-350-24	12 V PSU ⚠ model	buck 12 → 6 V	PSU bracket	M4 rivnut, Ø6 hole
B13	B14	B15	P03	B28
have ☐	have ☐	have ☐	have ☐	have ☐

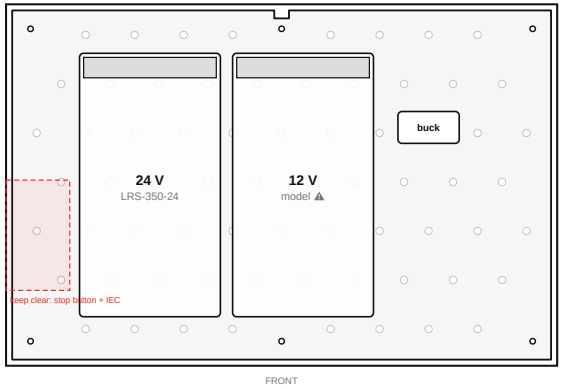


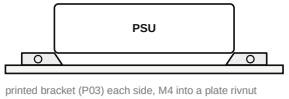
8x
M4 × 8 ⚠ datasheet
B31
have ☐

1. Read both PSU labels. Write the 12 V model and its rated current in the doc (Open questions 2).
2. Read the buck's chip: LM2596 (2–3 A) or XL4015 (5 A).
3. Stand the PSUs on the plate as drawn, terminal strips facing back. Mark the bracket holes on the grid, set M4 rivnuts, bolt down.
4. Keep the red zone clear: the stop button reaches about 51 mm in behind the IEC panel.

⚠ positions are the model's placeholders · P03 not modelled yet

⚠ PSU screw depth: check the datasheet

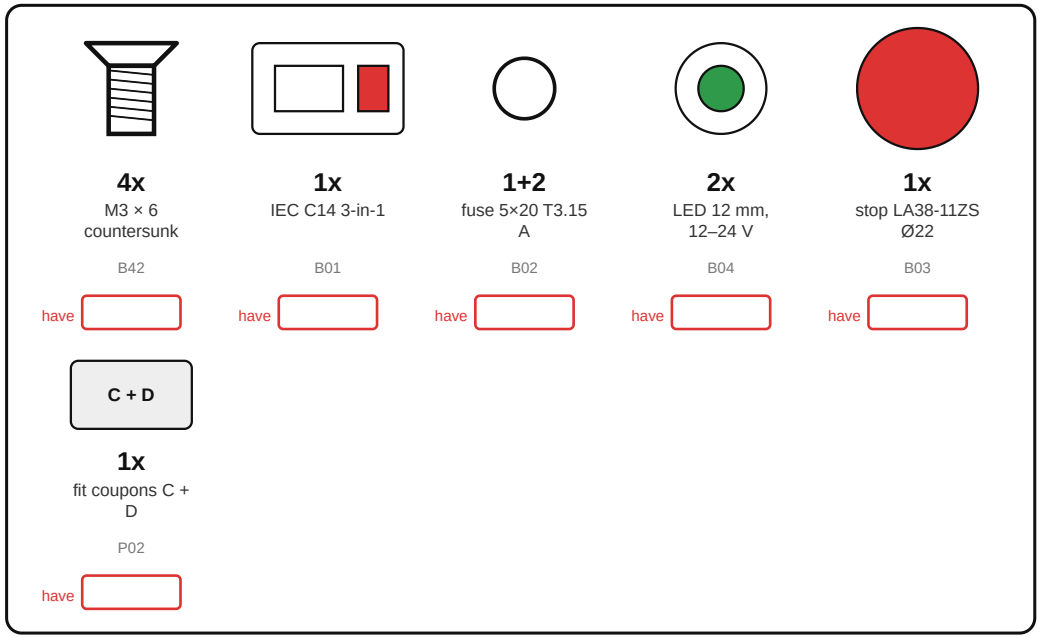




printed bracket (P03) each side, M4 into a plate rivnut

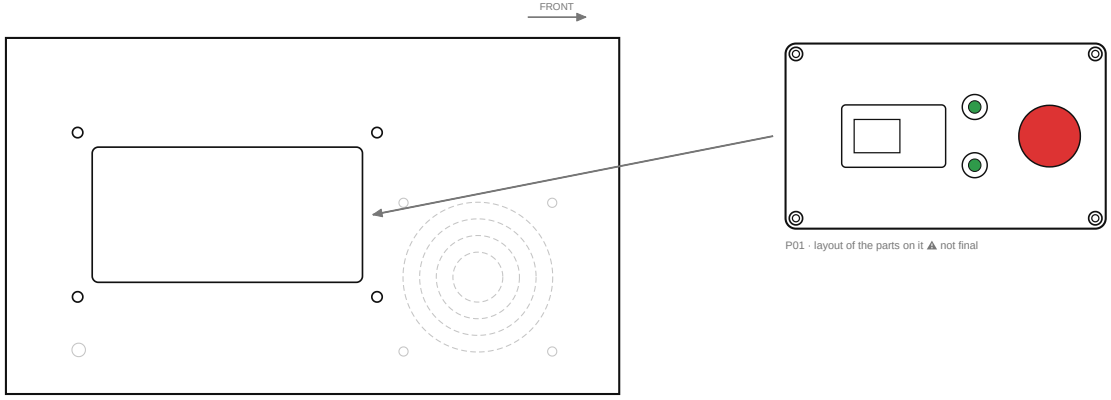
☑ **IEC module, stop button and fan clear the PSUs**

1.4 IEC panel



1. Print the fit coupons C and D first (P02). Pick the IEC frame that snaps in firmly; note the LED and stop hole sizes.
2. Measure the module's tabs (4.8 or 6.3 mm) and the stop button's AC rating before buying spades.
3. Into the panel: IEC module (snaps into the 1.5 mm thin zone), 2 LEDs, stop button. Fuse T3.15 A in the drawer.
4. Panel onto the wall, lip into the opening, 4 countersunk M3 into the rivnuts.

⚠ P01 not modelled yet: drawn from the steel and the parts list

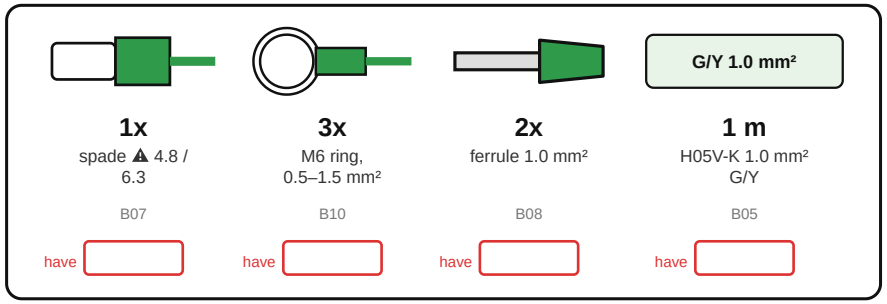


Power box, left wall, from outside

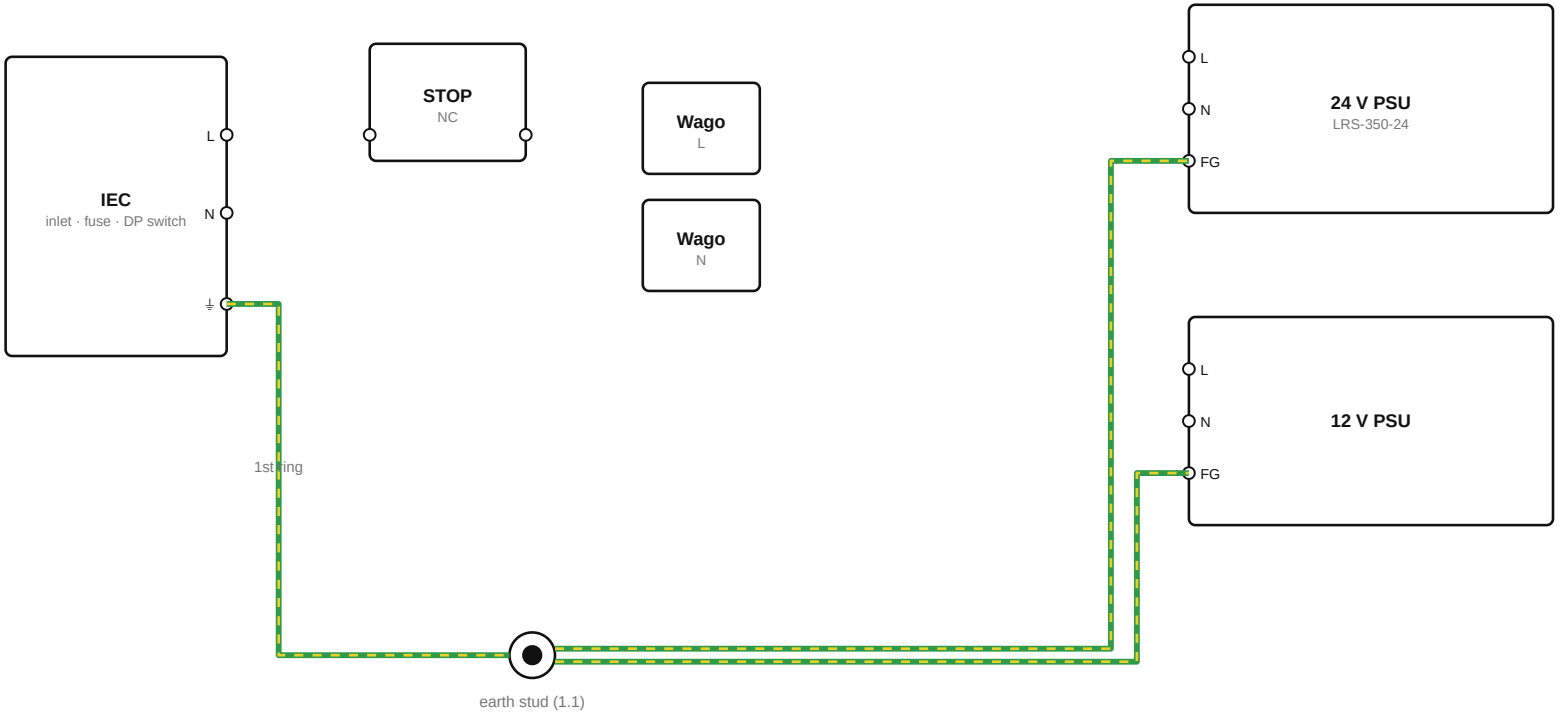
P01 · layout of the parts on it ⚠ not final

☑ **Lip seats · heads flush · stop latches and twists out**

1.5a Mains wiring · earth



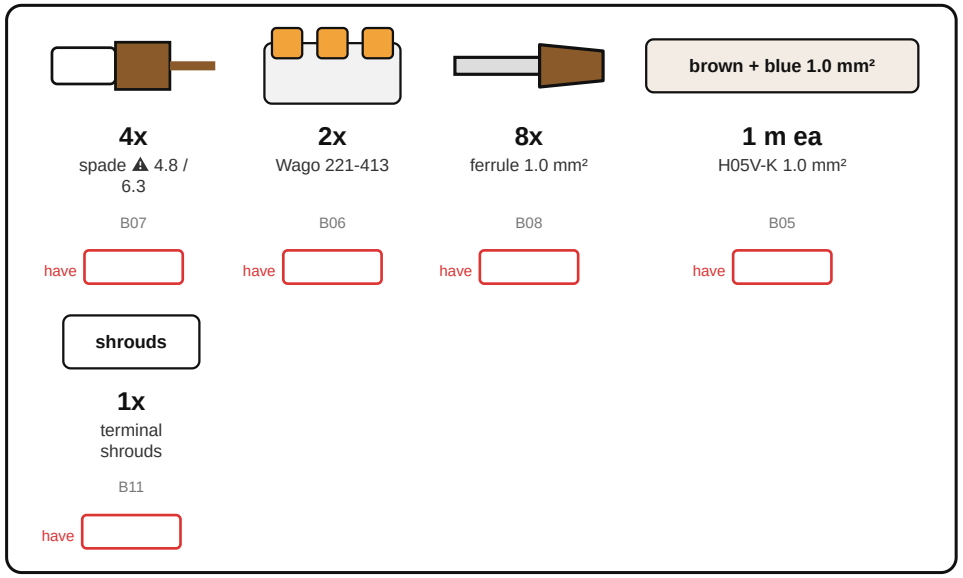
1. Unplugged. Earth first: IEC earth tab → the stud, on the first ring, own nut.
2. Stud → FG of the 24 V PSU, and stud → FG of the 12 V PSU: one wire each, on the second ring stack.
3. Leave each earth wire a little longer than L and N will be, so it pulls tight last.



☑ **Beep: IEC earth pin ↔ stud ↔ both PSU cases ↔ body**

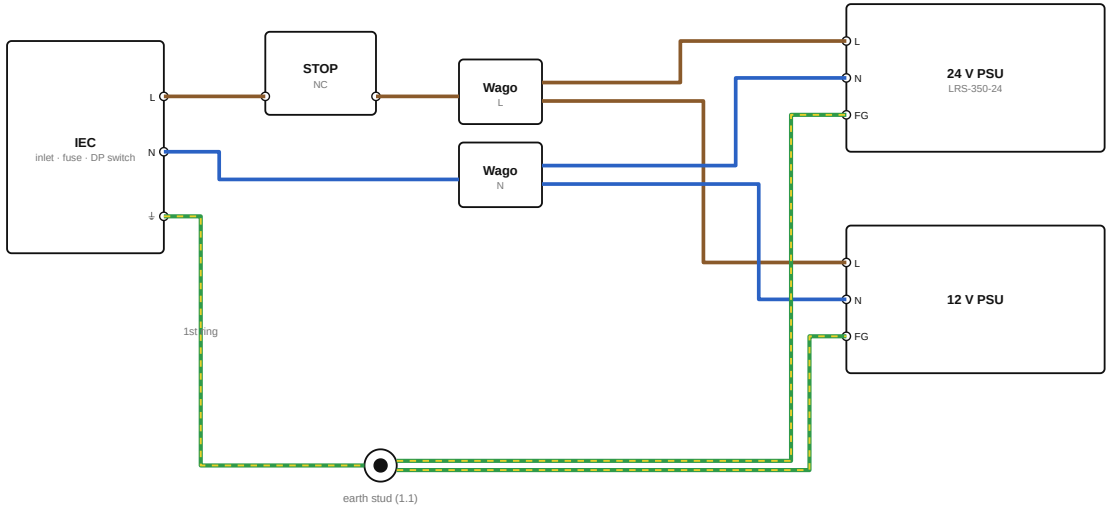
1.5b

Mains wiring · live and neutral



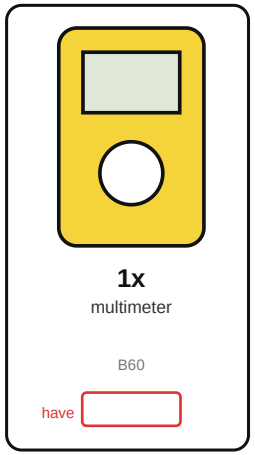
1. Live: IEC switched L → stop button NC → Wago L → L of each PSU.
2. Neutral: IEC switched N → Wago N → N of each PSU.
3. Ferrules on every end under a screw; spades on the IEC tabs. Shroud the IEC tabs and both PSU input strips.
4. Tie the mains bundle to the left half of the plate, away from where DC will run.

⚠ the stop button breaks L only: the IEC rocker is the double-pole switch

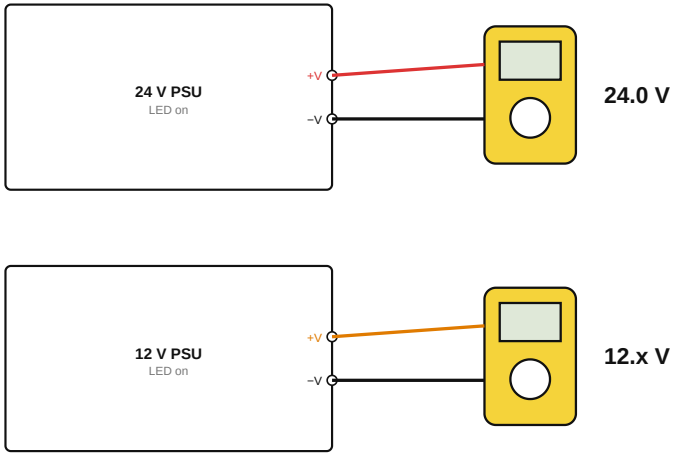


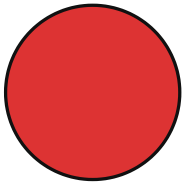
☑ **No beep L↔earth or N↔earth, rocker on and off**

1.6 First mains power-on



1. Nothing on the DC side. Lid off, hands clear. Plug in, rocker on.
2. Rocker lights; both PSU LEDs on. Meter 24 V and 12 V at the outputs. Trim the 24 V to 24.0 V.
3. Press the stop: both go dark. Twist to release: both back.
4. Rocker off, unplug. Wait for the PSU LEDs to fade before touching anything.





press: both dark
twist: both back

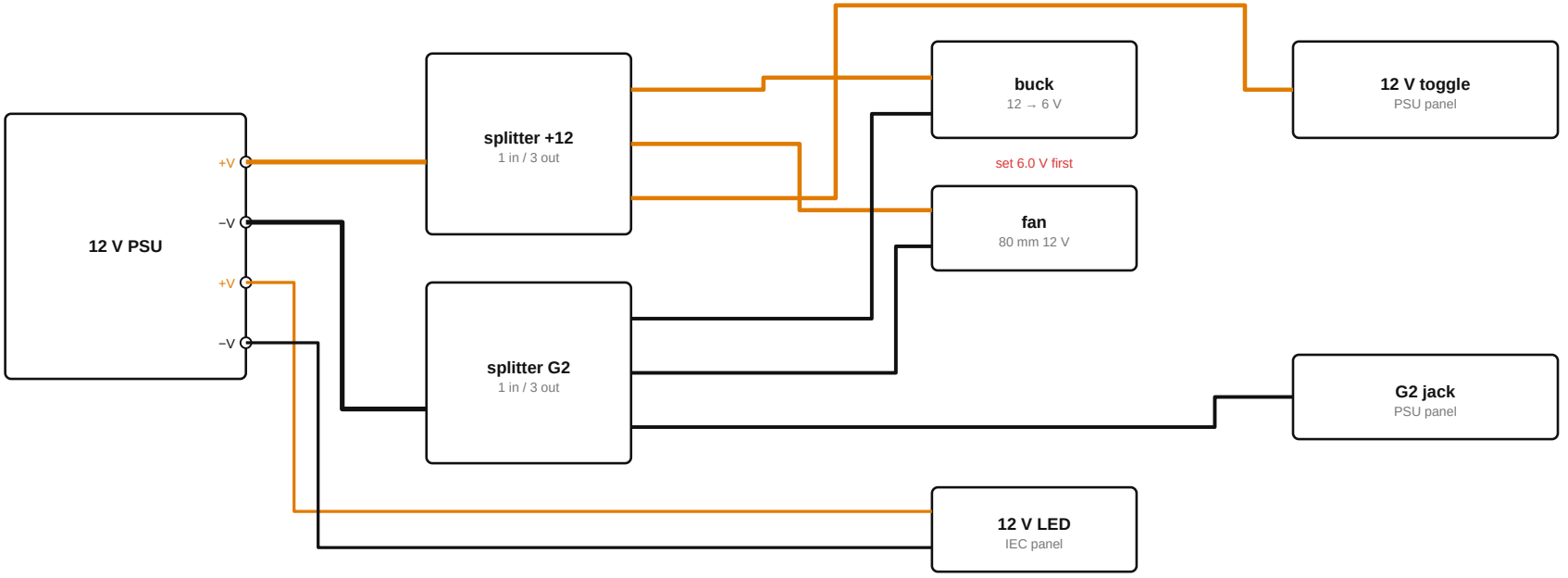
☑ **24.0 V and 12 V present · stop cuts both**

1.7a DC · 12 V and 6 V

PCT 2×6	0.75 mm²	
1x PCT 2×6 splitter	**3 m ea** 0.75 mm² silicone	**12x** ferrule 0.75 mm²
B16	B18	B08
have ☐	have ☐	have ☐

1. 12 V PSU +V → splitter +12 → buck IN+, fan +, the 12 V toggle.
2. 12 V PSU −V → splitter G2 → buck IN−, fan −, the G2 jack. One 0 V wire to the buck: its IN− and OUT− are the same copper.
3. 12 V LED straight off the PSU's second +V / −V pair.
4. Meter on the buck's output, nothing connected: set 6.0 V.

⚠ wire size: confirm against the 12 V PSU's label



LED from the PSU's spare +V / −V screws ⚠ check it has two pairs

☑ **Buck reads 6.0 V unloaded**

1.7b DC · 24 V

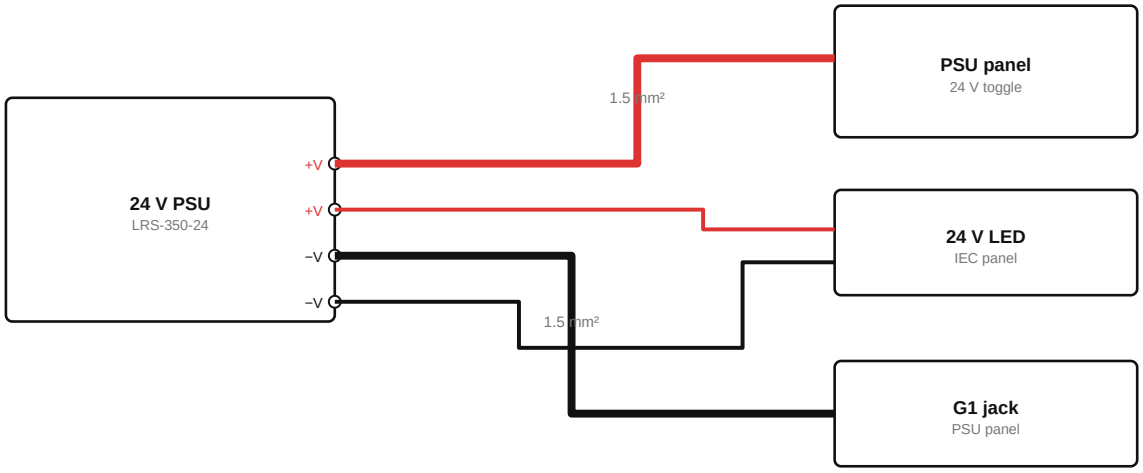
1. 24 V PSU +V → the 24 V toggle. −V → the G1 jack.
2. 24 V LED off the PSU's second +V / −V pair.
3. 1.5 mm² throughout on 24 V and G1: 16 AWG is too thin.

1.5 mm² red + black

3 m ea
1.5 mm² (not 16 AWG)
B17
have



6x
ferrule 1.5 mm²
B08
have

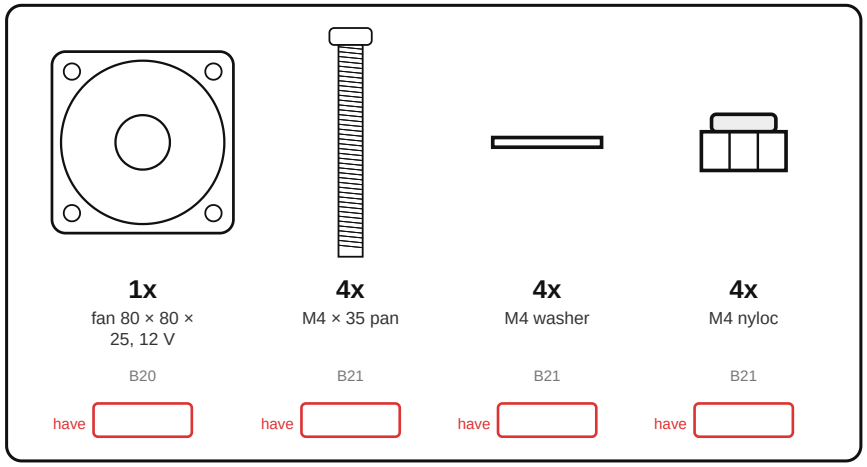


G1 and G2
are NOT joined
in this box

star point: Open questions 4

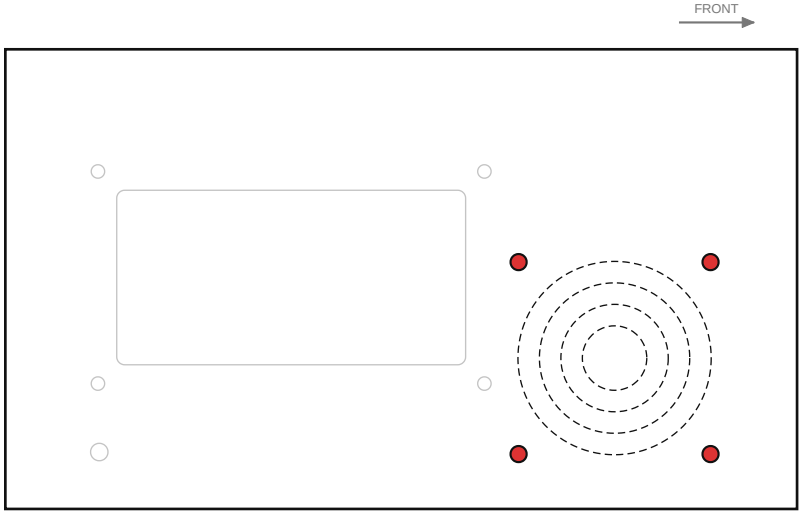
☑ **Unplugged: G1↔G2 open · G1, G2 ↔ stud open**

1.8 Fan · power box

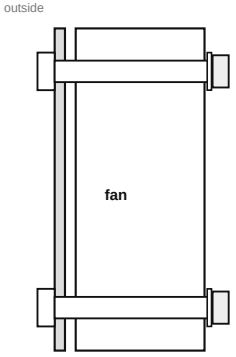


1x	4x	4x	4x
fan 80 × 80 × 25, 12 V	M4 × 35 pan	M4 washer	M4 nyloc
B20	B21	B21	B21
have ☐	have ☐	have ☐	have ☐

1. Fan inside, against the grille. Airflow direction: see Open questions 6 (arrow on the fan's side).
2. M4 × 35 from outside, washer and nyloc inside. Snug, don't crush the fan frame.
3. Lead: + to 12 V, − to G2 (1.7a).

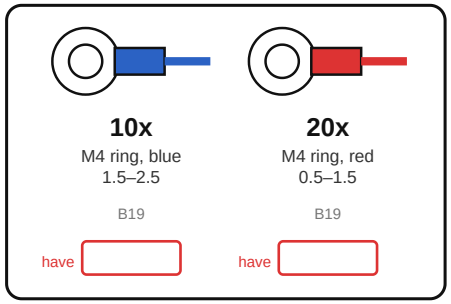


Power box, left wall, from outside

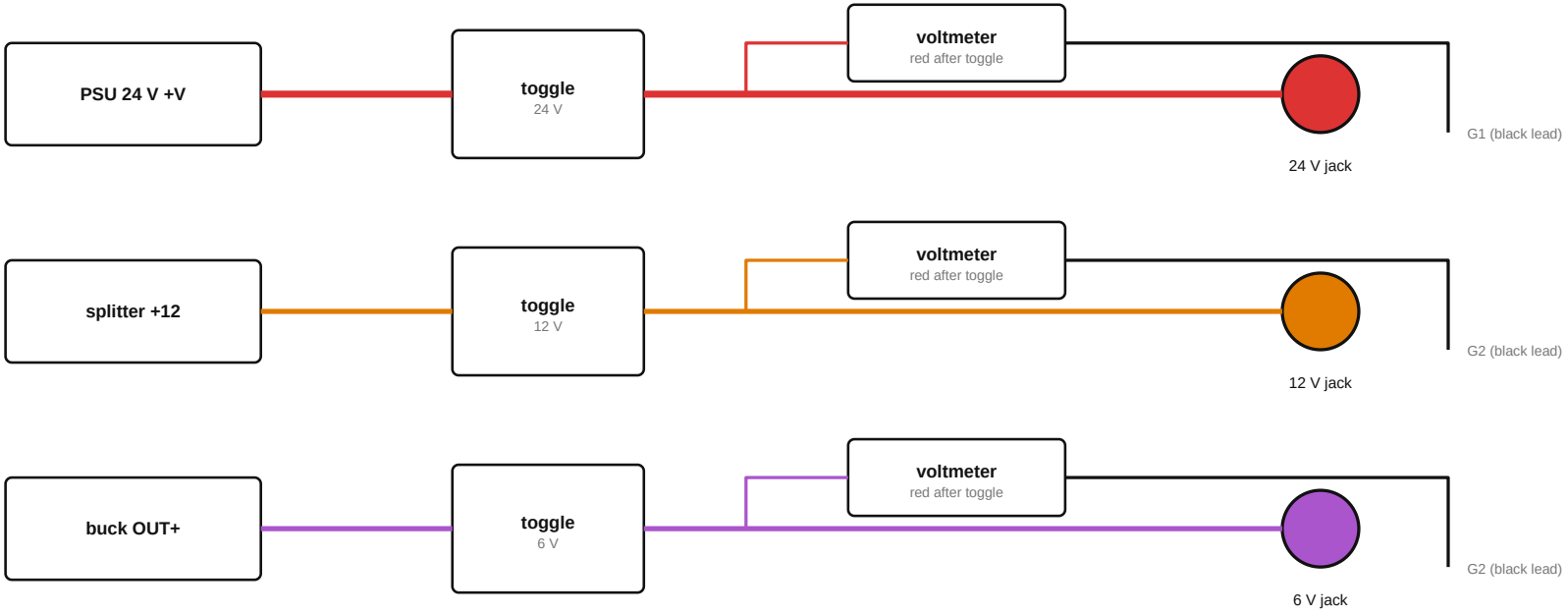


☑ **Blades turn freely by hand**

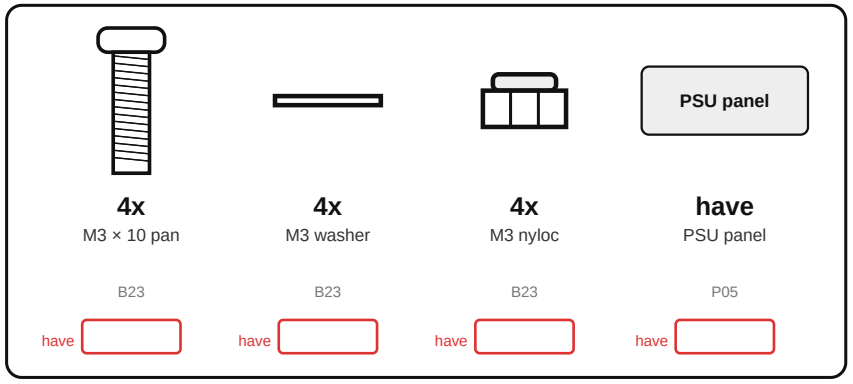
1.9a PSU panel · wiring



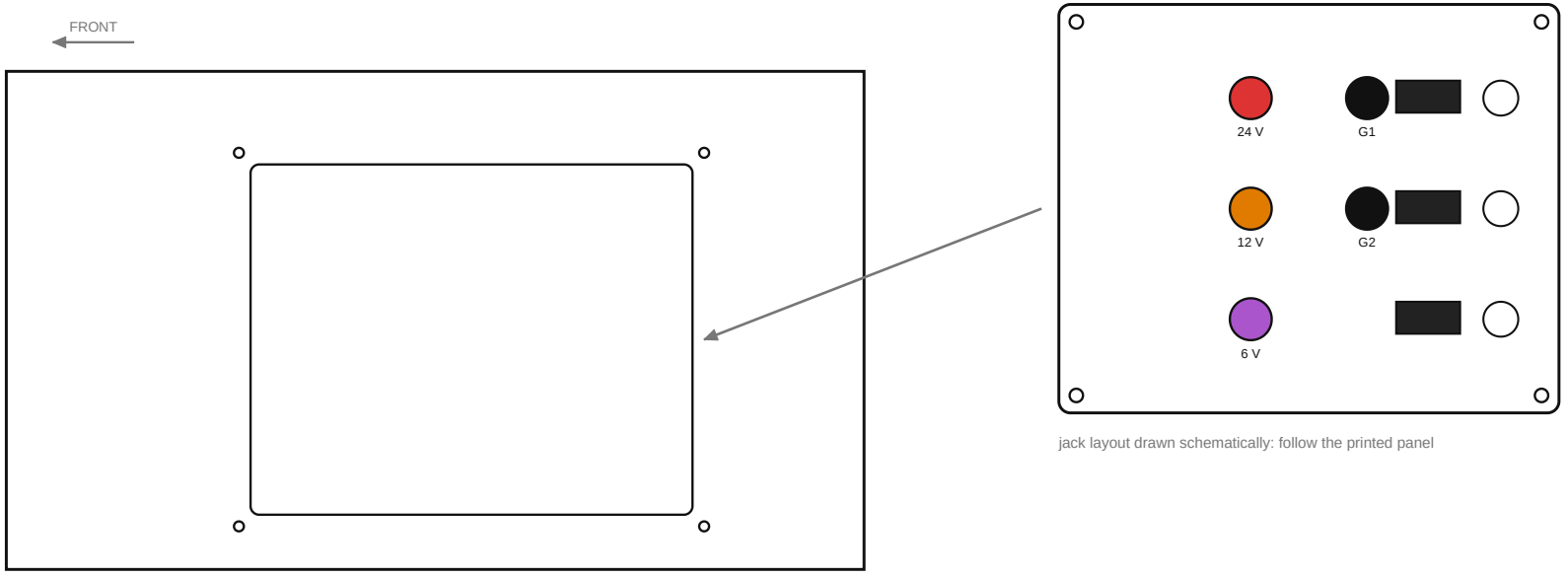
1. Each rail: source → toggle → jack.
2. Voltmeter red lead after the toggle (it doubles as the rail-on light); black lead to that rail's own ground.
3. Rings on the jack studs: blue (1.5–2.5) for 24 V and G1, red (0.5–1.5) for 12 V, 6 V, G2.



1.9b PSU panel · mount



1. Wire the panel on the bench first (1.9a), leaving 15 cm of slack into the box.
2. Panel onto the right wall: M3 × 10 from outside through the Ø3.4 holes, washer and nyloc inside.

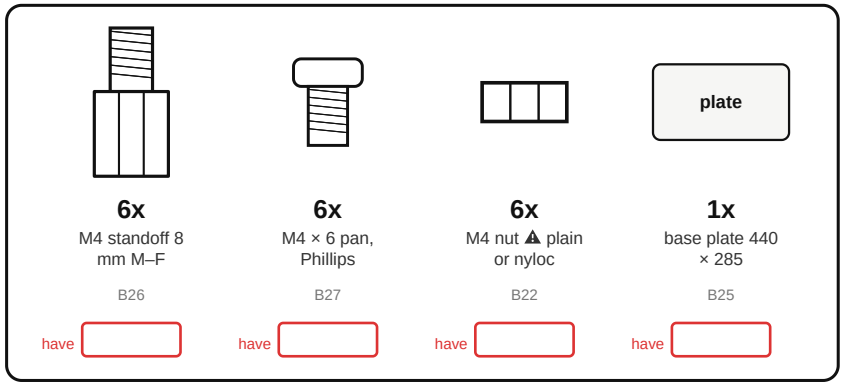


Power box, right wall, from outside

jack layout drawn schematically: follow the printed panel

☑ **Powered: each toggle shows its volts · stop: all dark**

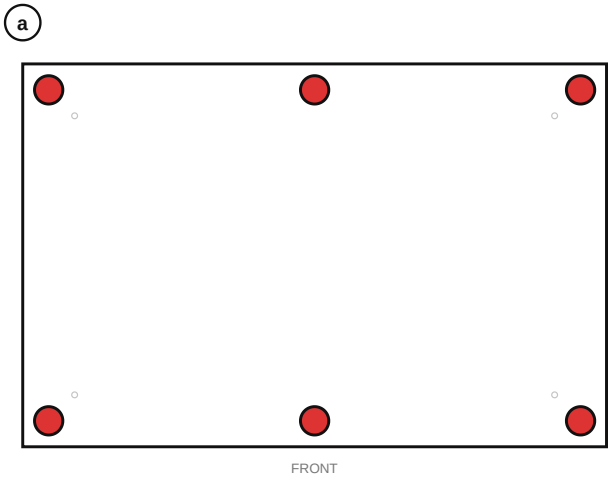
2.1 Base plate · control box



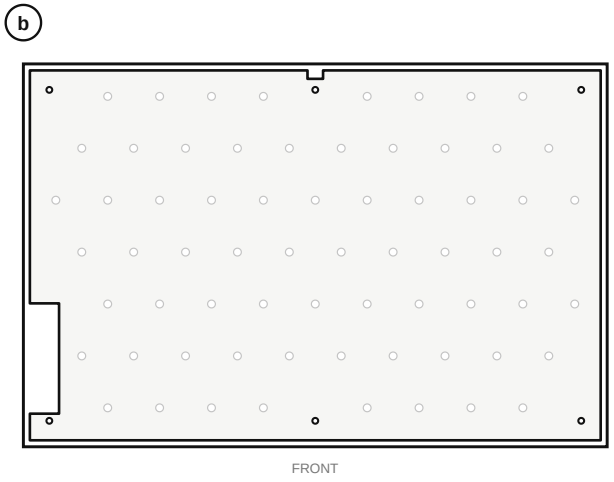
1. a Six standoffs onto the floor: M4 × 6 pan screw up from underneath into the female end.
2. b Plate on, studs through its standoff holes, a nut on each. The plate comes out from inside by undoing 6 nuts. File a 12 × 6 mm notch in the plate's front edge at x +125 first: the window's bottom rivnut needs it.

Plates: 2 mm mild steel, already cut

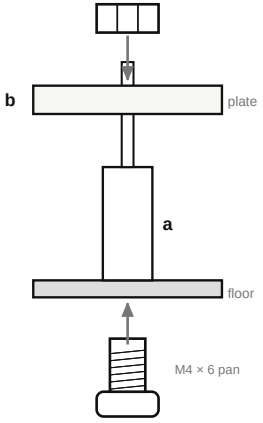
⚠ notch: at 8 mm the window rivnut overlaps the plate by 0.2 mm



Control box floor, from above

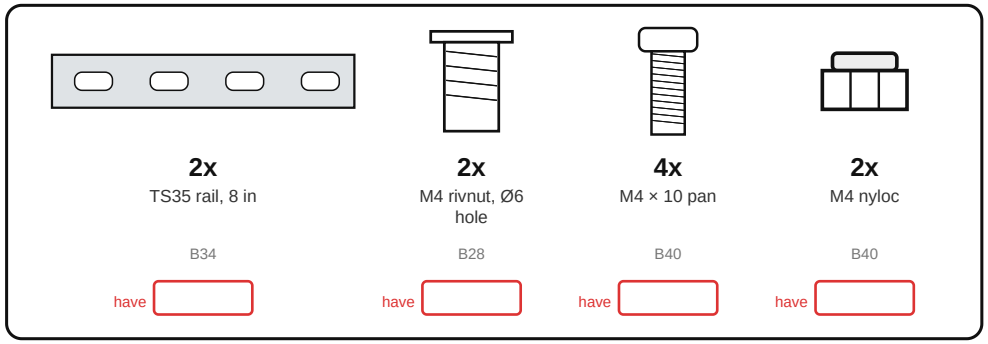


each of 6



☑ **Plate doesn't rock · no screw head below the feet**

2.2 DIN rails

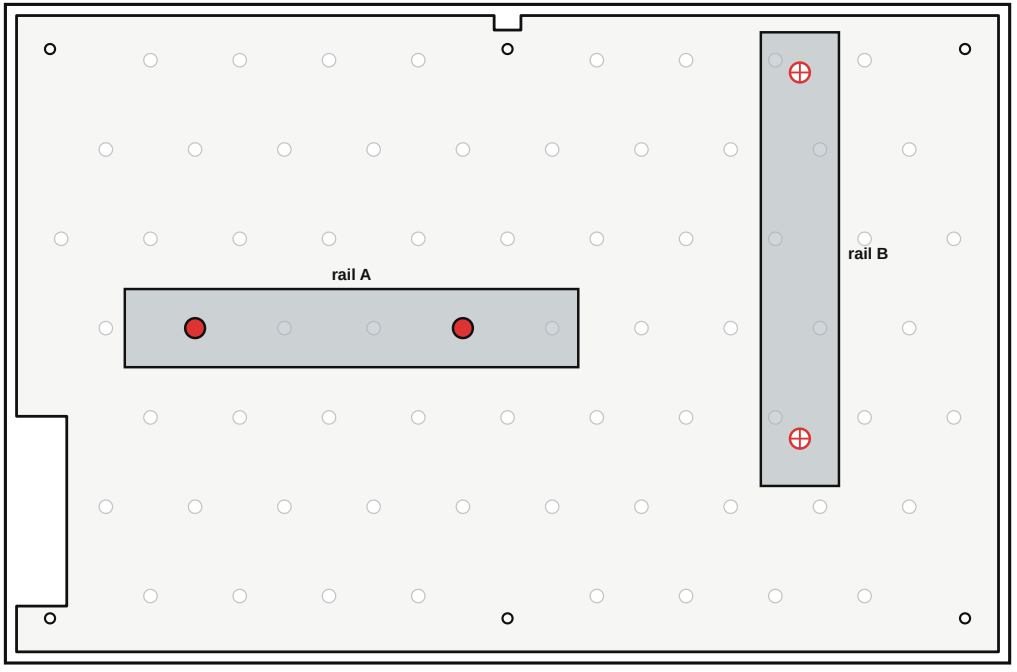


2x	2x	4x	2x
TS35 rail, 8 in	M4 rivnut, Ø6 hole	M4 × 10 pan	M4 nyloc
B34	B28	B40	B40
have ☐	have ☐	have ☐	have ☐

1. Rail A (drivers): along the box on grid row z −2.5. M4 rivnuts in the grid holes at x −140 and −20.
2. Rail B (board): front to back at x 131. Drill Ø4.5 at z −117 and +47; M4 bolt, nyloc under the plate.
3. Lay each rail on the plate and mark through its slots before setting or drilling anything.

⚠ rail slots must land on these holes: mark first

Moves rail A 12.5 mm forward of the layout check: still 51 mm clear of the GX zone



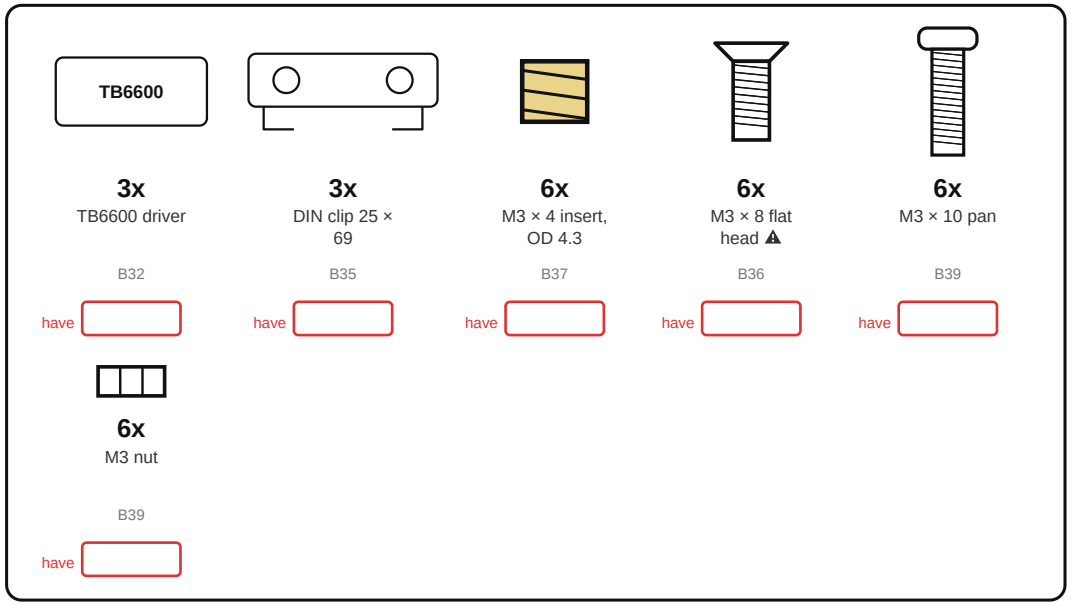
FRONT

M4 rivnut in a grid hole

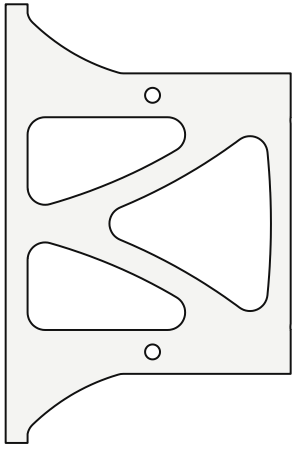
drill Ø4.5, M4 + nyloc

☑ **Both rails flat; don't move when pushed**

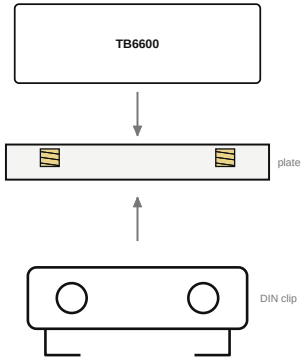
2.3a Driver plates



1. Print 3 TB6600 plates at the corrected 89 mm hole pitch (P07).
2. Melt the M3 inserts into the clip bosses. Clip under each plate, countersink on the rail side, M3 flat heads.
3. Driver onto its plate: M3 × 10 + nut through both flanges.

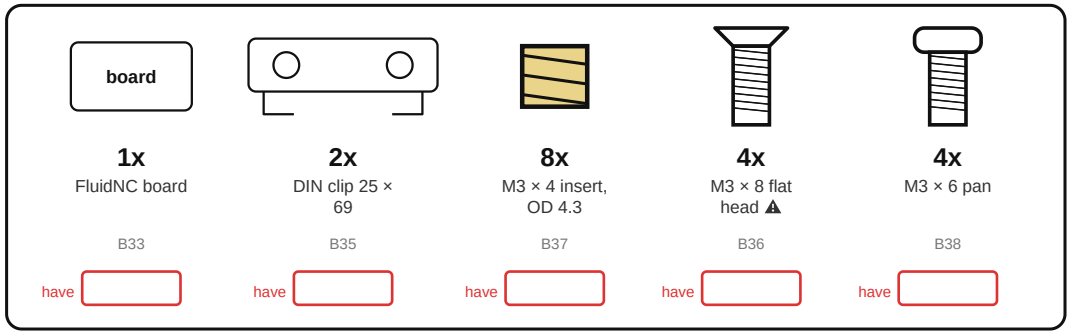


printed plate, from above (traced from its STL)



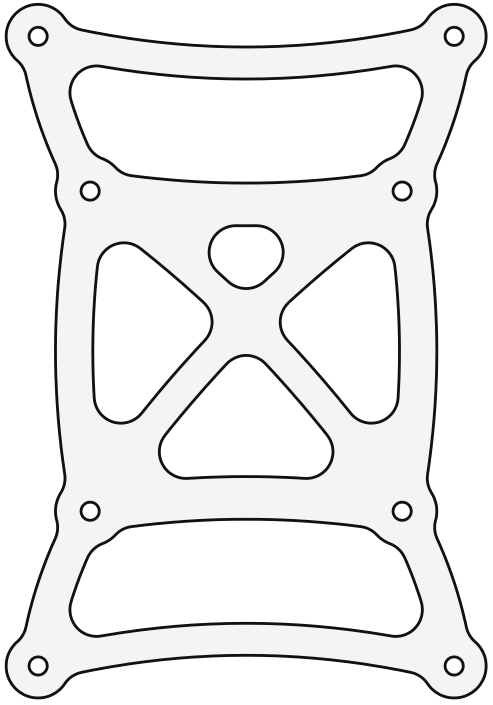
☑ **Each plate snaps onto a rail offcut and releases**

2.3b Board plate

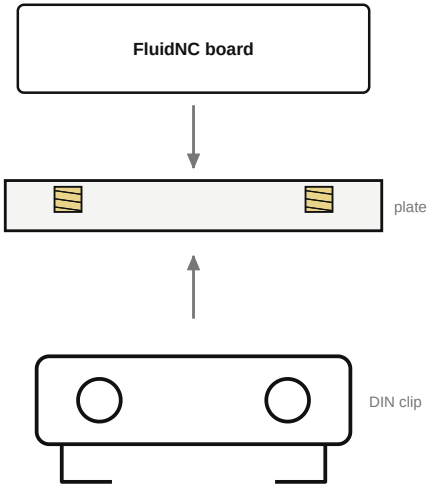


1. Print the board plate with the rotated clips (P08).
2. Inserts into the clip bosses and the 4 board bosses. Clips under, M3 flat heads.
3. Board on with 4 × M3 × 6: USB to the back, driver signals left, I/O right, power to the front.

⚠ board hole spacing 78 × 118: best-effort measurement



printed plate, from above (traced from its STL)

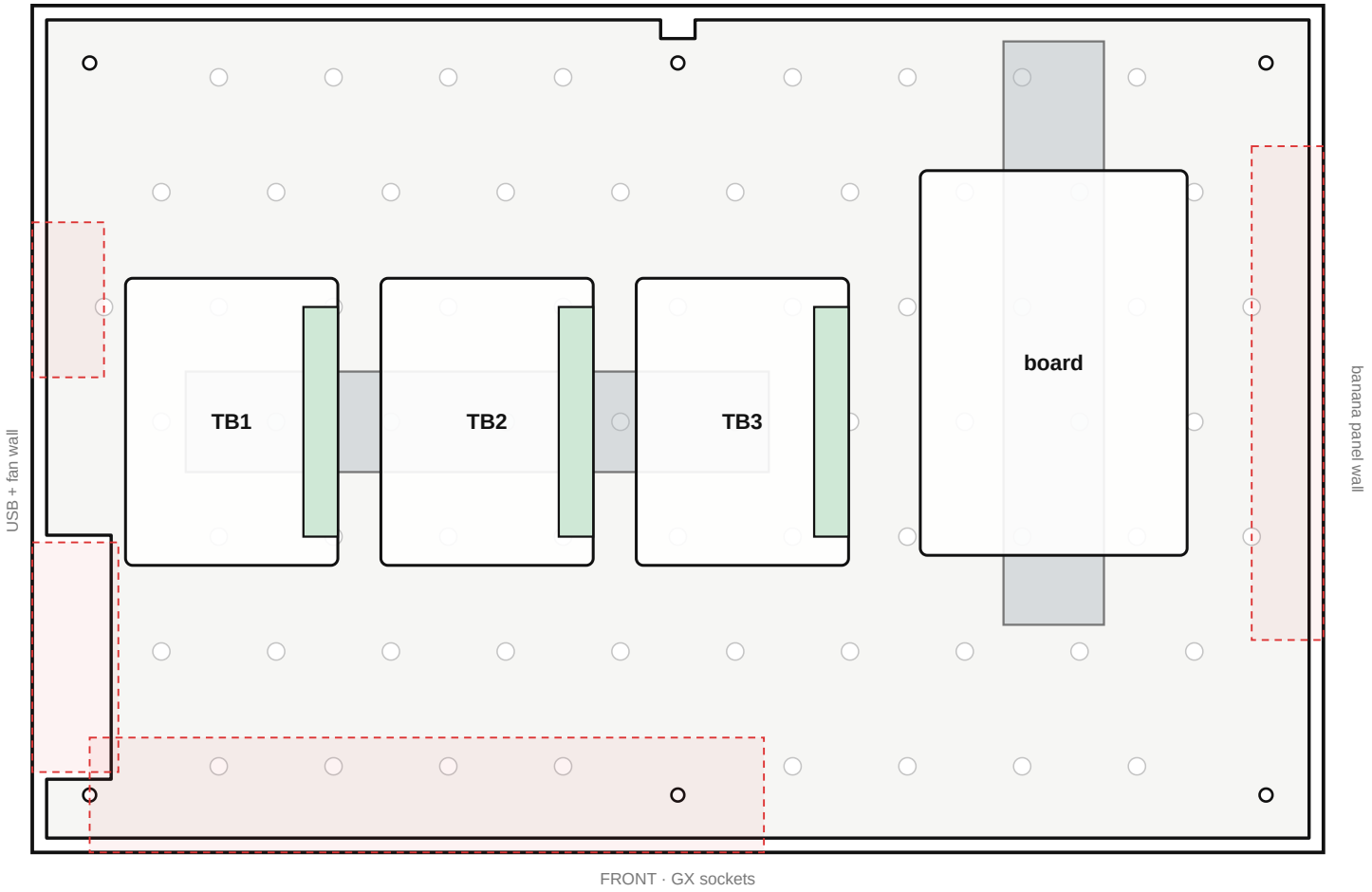


☑ **PCB sits flat, nothing touches underneath**

2.4 Plates onto the rails

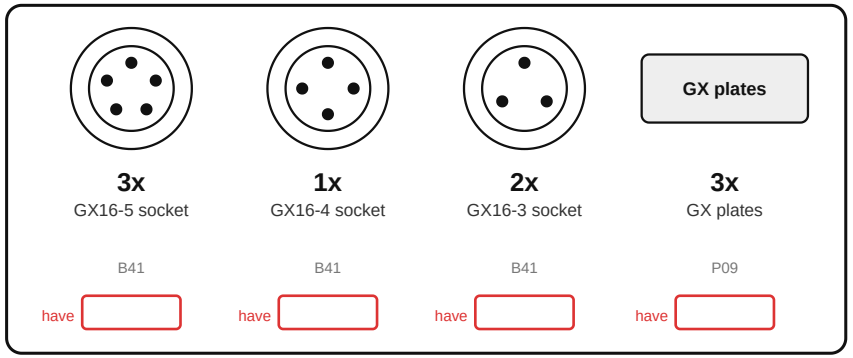
1. Snap the three TB6600 plates onto rail A, 89 mm apart: about 7.5 mm spare at each end.
2. Board plate onto rail B.
3. Red dashed: keep-outs for the fan, panels and GX plugs. Nothing goes in them.

⚠ the model warns: driver TB1 sits 1.5 mm from the GX sockets; nudge right if tight



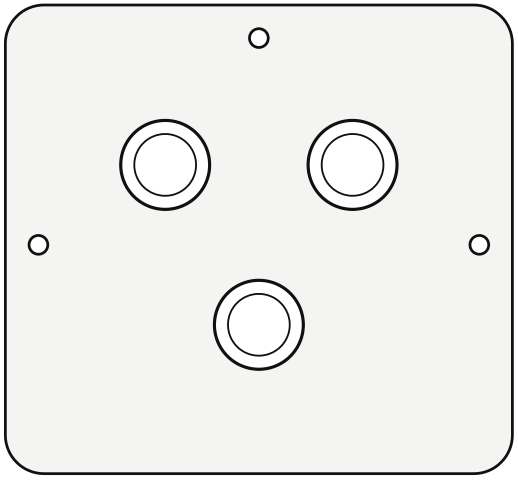
☑ **About 51 mm clear above the drivers to the lid**

2.5a GX sockets into the plates

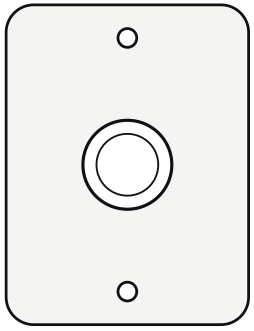


1. Print the three GX plates (P09).
2. Sockets into the plates from the front, nut behind, tight. The nut seat stops them turning.
3. Solder the socket tails now, on the bench (pin schedule 2.9b). Then the plates go on.

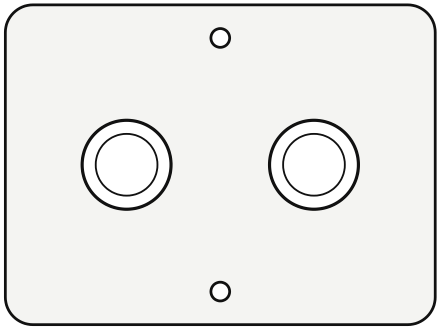
Different pin counts on purpose: nothing mates the wrong socket



steppers · GX16-5 × 3



servo · GX16-4 × 1



endstops · GX16-3 × 2

☑ **Each plug mates and locks**

2.5b

GX plates onto the front

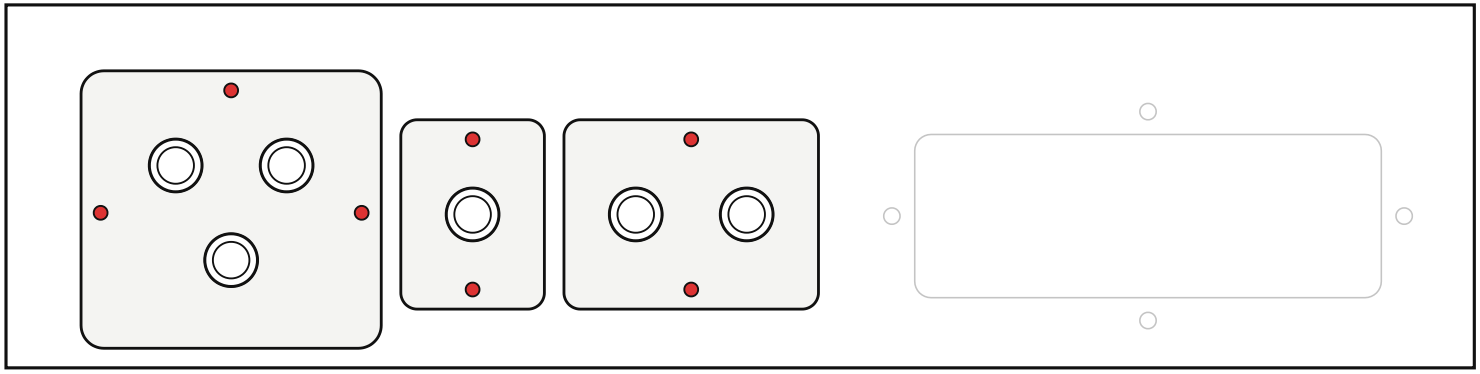


7x
M3 × 6
countersunk

B42

have ☐

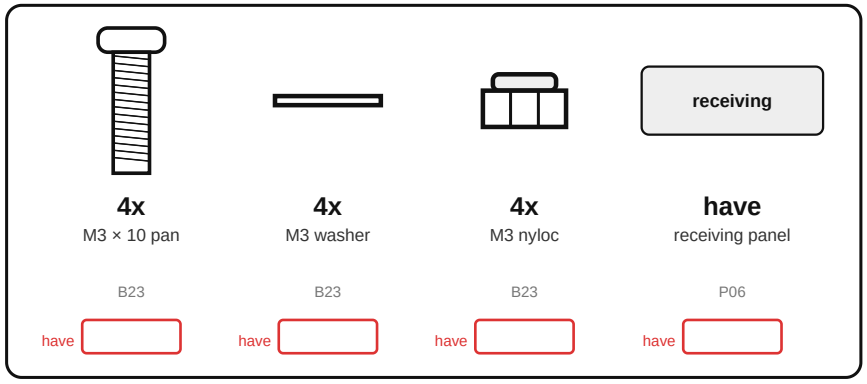
1. Feed the tails in through the openings first.
2. Plate on, lip into the opening, countersunk M3 × 6 into the rivnuts (red): 3 + 2 + 2.



Control box front, from outside

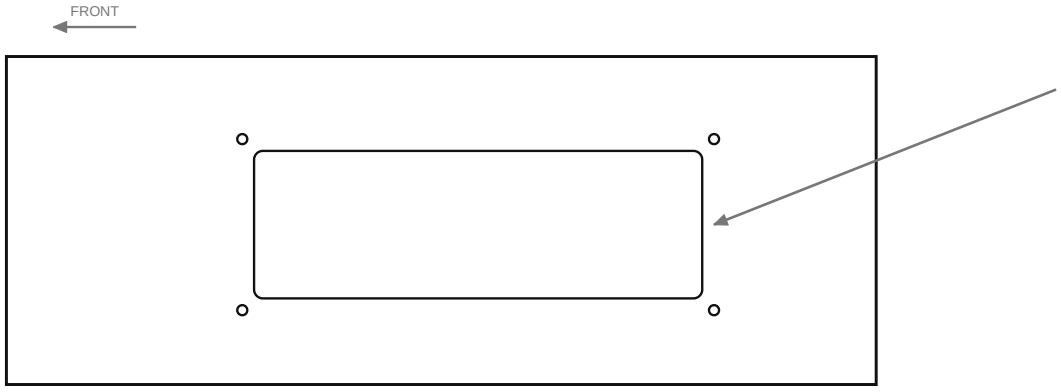
☑ **Backshells tighten without fouling each other**

2.6 Receiving panel

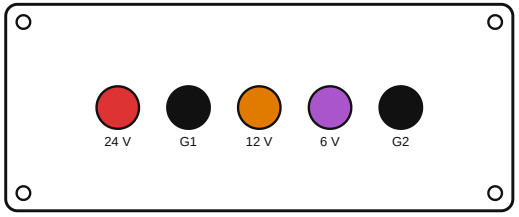


1. Rings on the 5 posts first (2.8), then the panel on.
2. M3 × 10 from outside through the Ø3.4 holes, washer and nyloc inside.

⚠ post order drawn as 24 V, G1, 12 V, 6 V, G2: follow your printed panel's labels



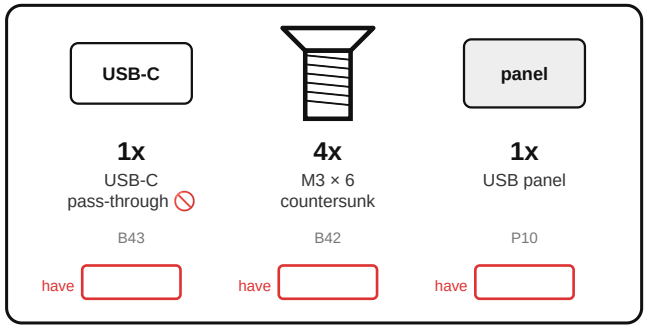
Control box, right wall, from outside



jack layout drawn schematically: follow the printed panel

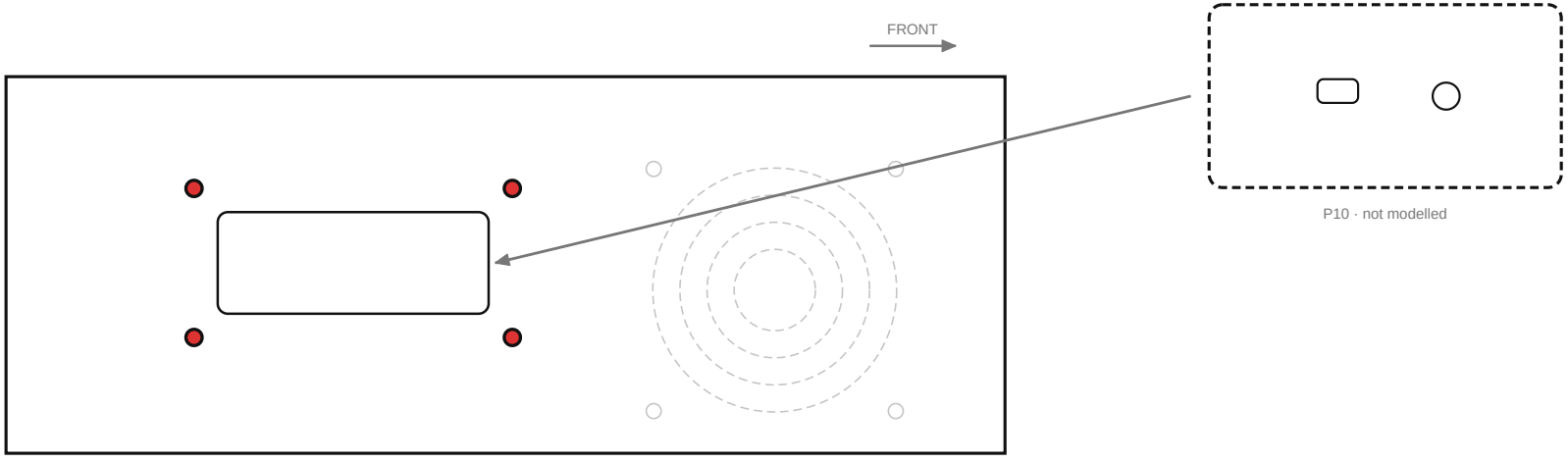
☑ **Posts don't spin**

2.7 USB / antenna panel



1. Decide the USB-C pass-through (B43) and whether the pendant socket goes in now (recommended: later).
2. Short USB lead from the panel to the board, along the board's back edge.
3. Panel on: countersunk M3 × 6 into the 4 rivnuts.

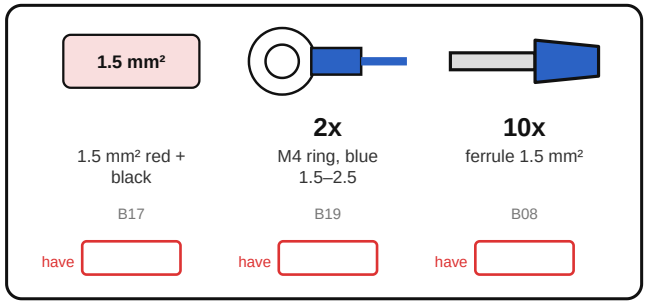
🚫 gate: USB-C part (B43) · pendant now or later



Control box, left wall, from outside

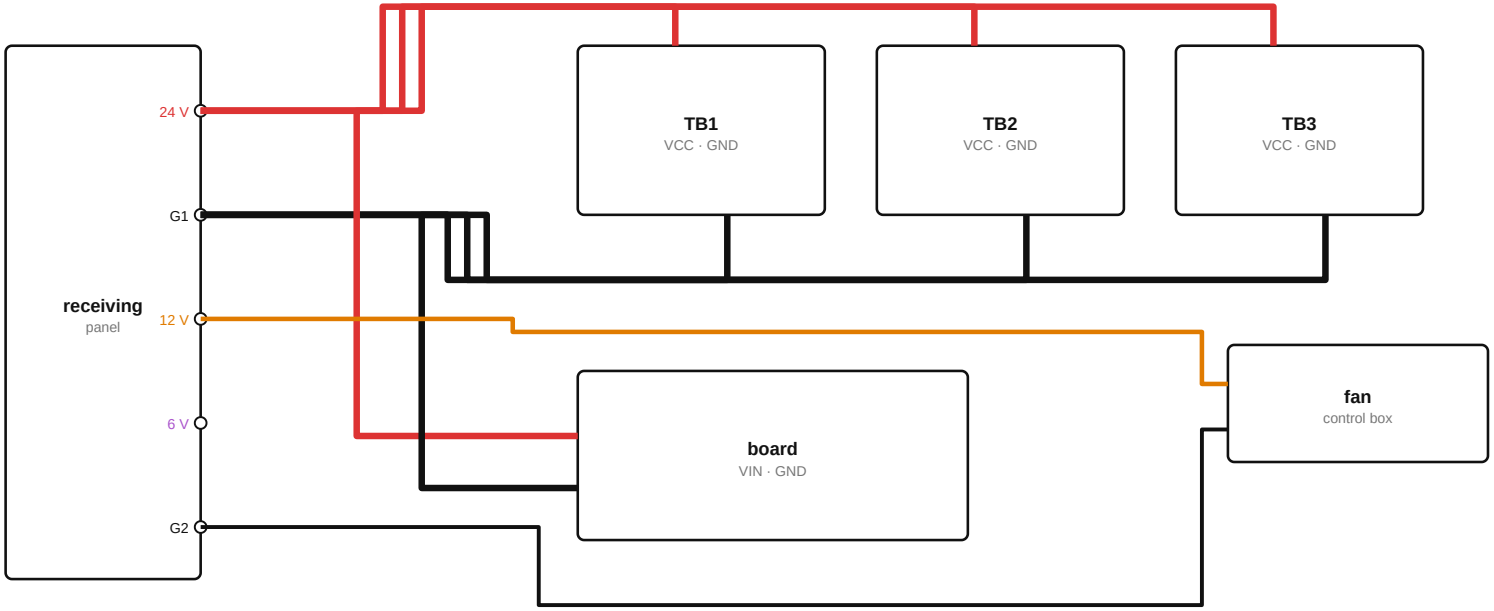
☑ **Laptop sees the board over the panel**

2.8a Power wiring · duct 1



1. 24 V post → VCC of each TB6600 and the board's VIN, 1.5 mm². G1 post → each GND. One wire per terminal; no daisy chains.
2. 12 V post → control fan +, G2 → fan −.
3. Before powering: read the board's 5 V regulator. A linear 7805 will cook on 24 V.

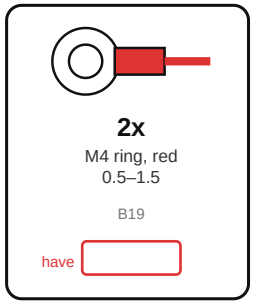
⚠ board regulator unverified (Open questions 11)



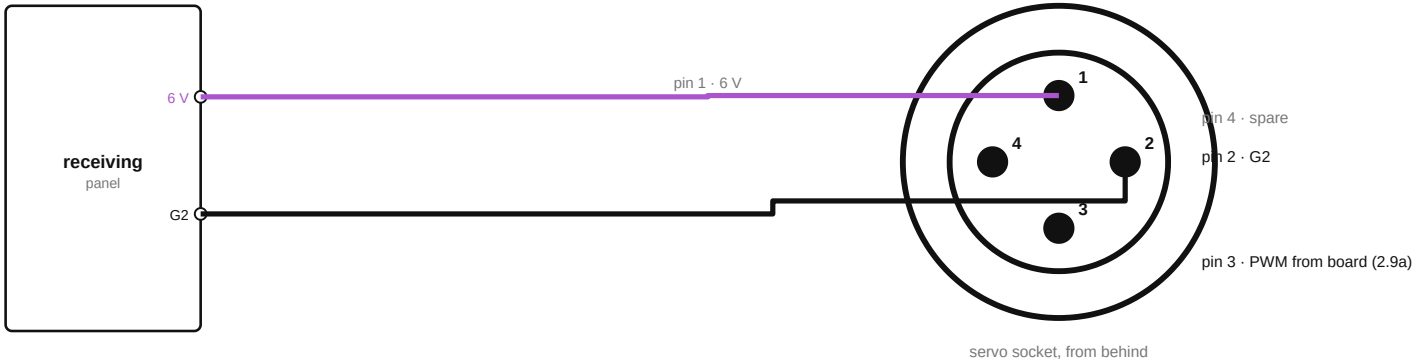
☑ **24 V ↔ G1 not a short · G1 ↔ body open**

2.8b

Power wiring · servo supply



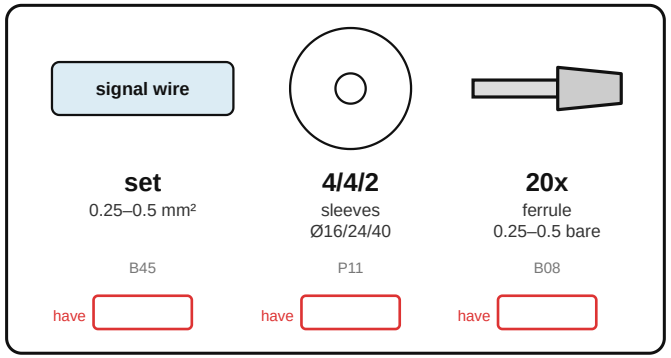
1. 6 V post → servo socket pin 1. G2 post → pin 2.
2. The board's PWM goes to pin 3 (2.9a). Pin 4 spare.
3. Fit the 2200 µF at the servo end, not here (B49).



Star point (Open questions 4): proposal is one short link G2 post → G1 post on this panel.
Needs your OK before it goes in.

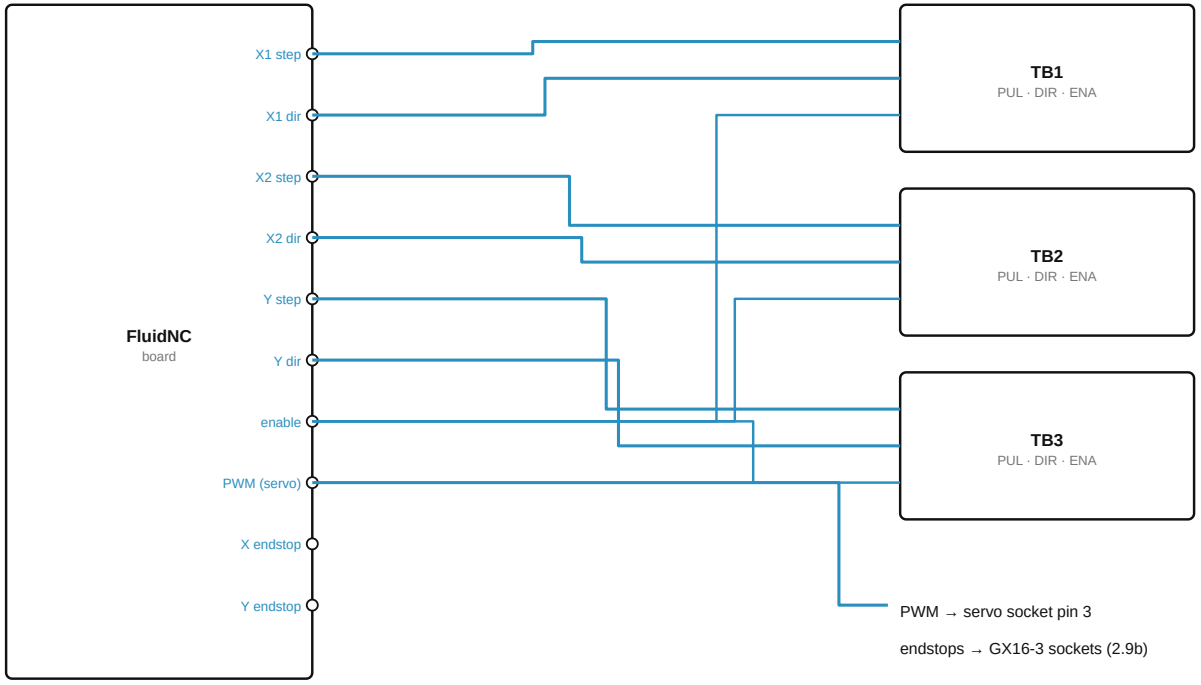
☑ **6 V post ↔ pin 1 beeps · pin 1 ↔ pin 2 open**

2.9a Signals · board to drivers

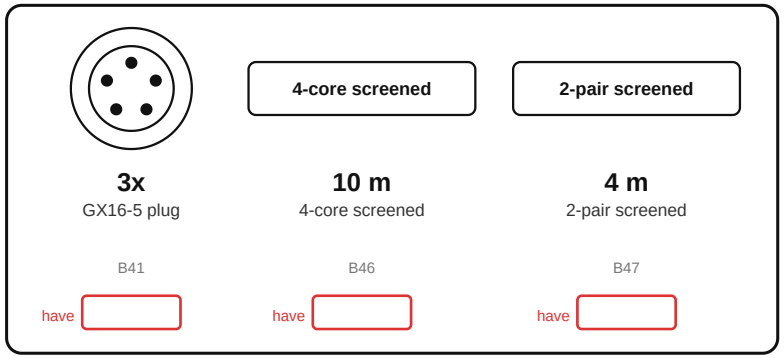


1. Step, dir and enable from the board's external-driver header to each TB6600.
2. Follow the board's pinout for PUL+ vs PUL− (common anode or cathode). Check each driver's silkscreen: clones reorder terminals.
3. 0.25–0.5 mm², labelled both ends. Bend with the jig sleeves (P11); guides (P15) along duct 2.

⚠ board output type and TB6600 terminal order unverified (Open questions 12)

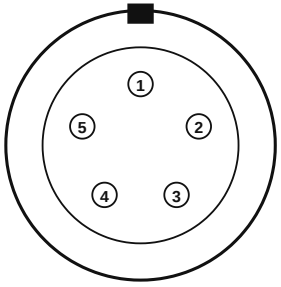


2.9b Signals · socket pins



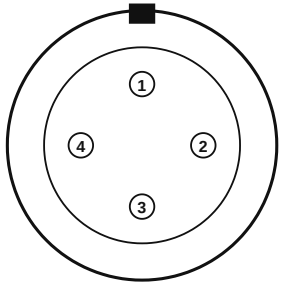
1. Motors: A+ A− B+ B− on pins 1–4; the cable screen on pin 5 → G1. Never earth a screen through the shell.
2. Find coil pairs with the meter: continuity = same coil. Don't trust Ender colours.
3. Endstops: screen at the box end only.

Seen from the solder side (behind the plate) ⚠ check the numbers moulded on your sockets



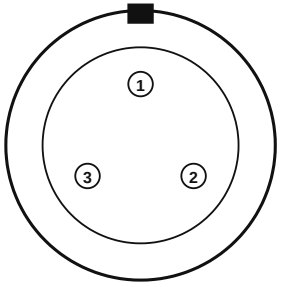
GX16-5 · steppers

1 A+
2 A−
3 B+
4 B−
5 screen → G1



GX16-4 · servo

1 6 V
2 G2
3 PWM
4 spare

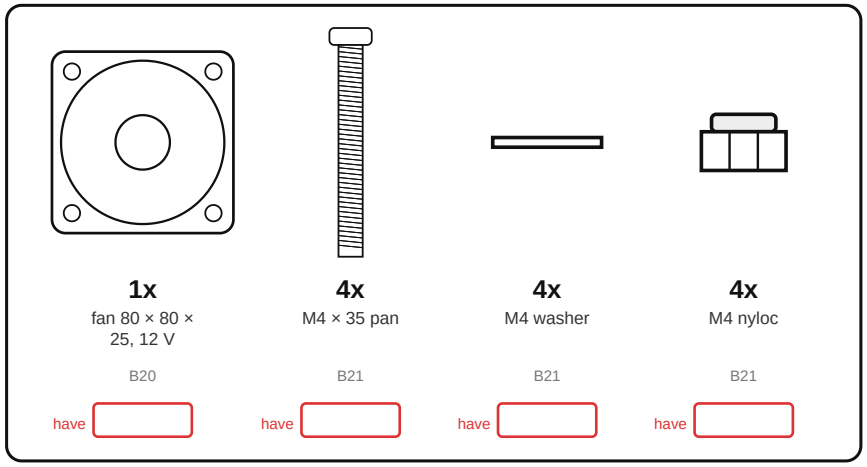


GX16-3 · endstops

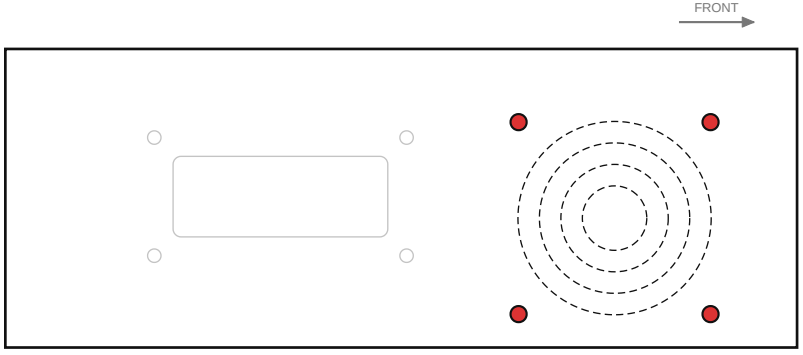
1 signal
2 return
3 screen (box end)

☑ **Continuity table ticked: every pin to its terminal**

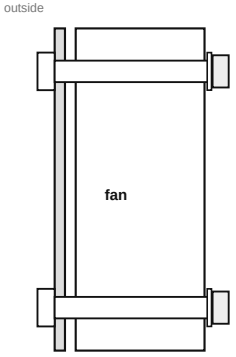
2.10 Fan · control box



1. Fan inside, against the grille. Airflow direction: see Open questions 6 (arrow on the fan's side).
2. M4 × 35 from outside, washer and nyloc inside. Snug, don't crush the fan frame.
3. Lead: + to 12 V, − to G2 (2.8).



Control box, left wall, from outside

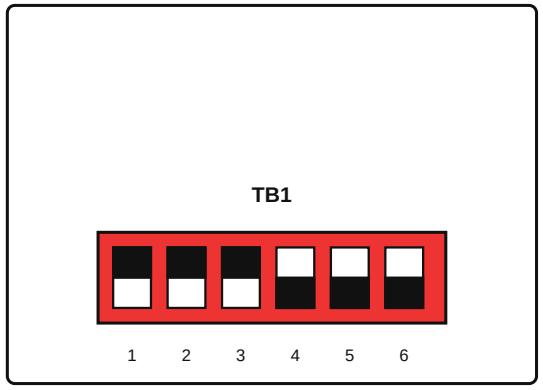


- [ ] **Blades turn freely by hand**

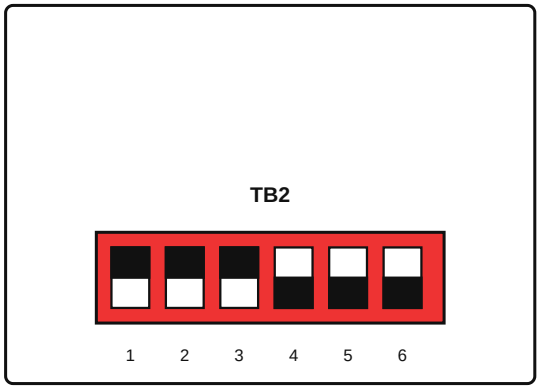
2.11 Driver switches

1. Current: the Ender 42-34 is about 0.84 A per phase. Use the nearest setting not above it; check whether the driver's table is peak or RMS.
2. Microstep: the same on all three, and it must match steps/mm in the FluidNC config.
3. Write the setting on each driver with a marker.

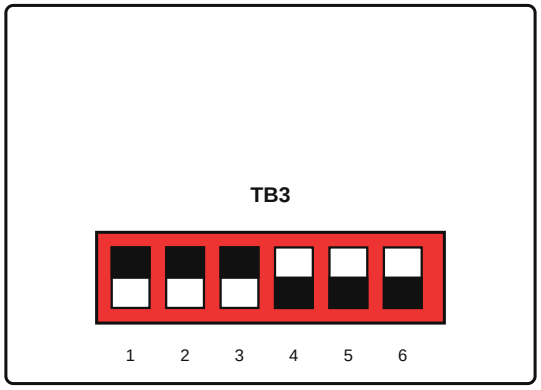
⚠ switch positions drawn are an example, not the setting



SW1–3: microstep · SW4–6: current



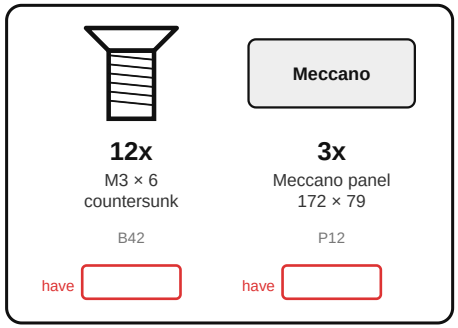
SW1–3: microstep · SW4–6: current



SW1–3: microstep · SW4–6: current

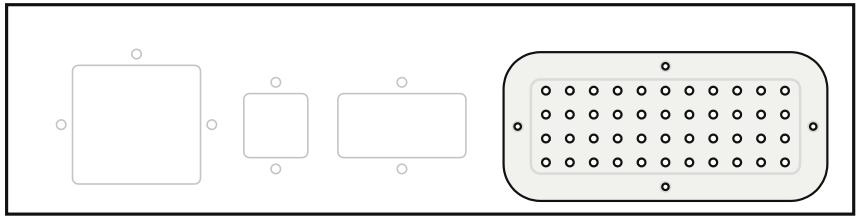
☑ **Same on all three · written on each**

3.1 Window panels

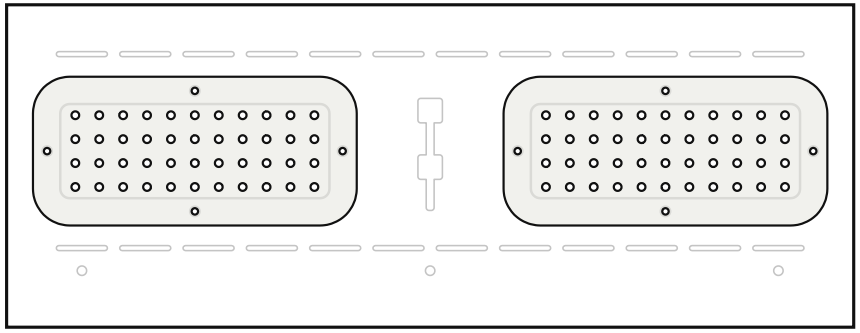


1 Print three Meccano window panels (P12): one per window slot file.

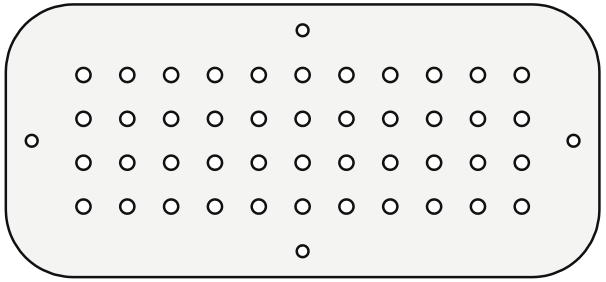
2 Lip into the opening, 4 countersunk M3 × 6 into the rivnuts.



control front, outside



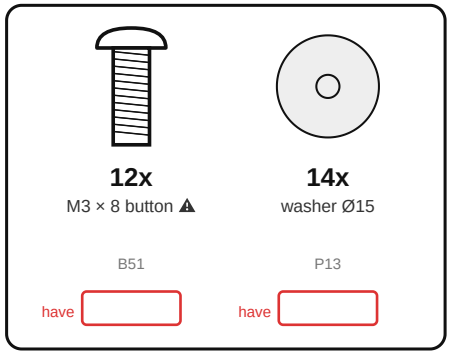
power front, outside



172 × 79, 12.7 mm Meccano pitch

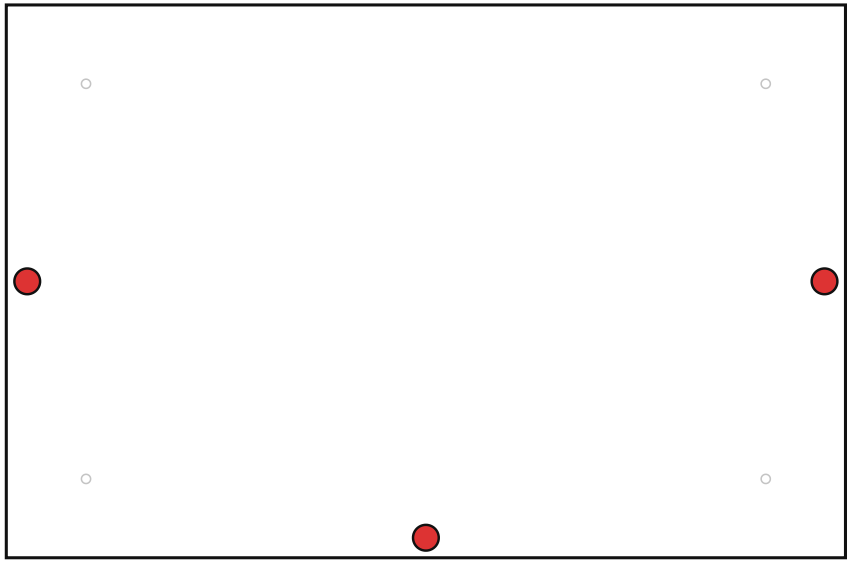
☑ **Panels flat, heads flush**

3.2 Lids

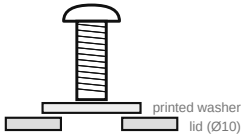


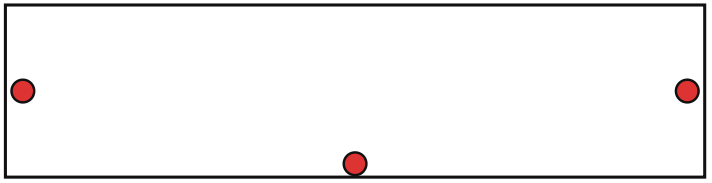
1. Power lid first. Ease the snug fit before forcing it (Open questions 13).
2. 6 screws per lid: 3 on top, 3 at the back. Button head M3 over a printed washer; the Ø10 holes clear the rivnut flanges.
3. Same for the control lid.

⚠ screw length · P13 not modelled yet



lid top, from above

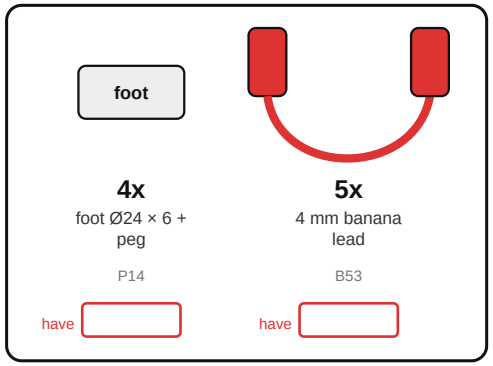




lid back, from behind

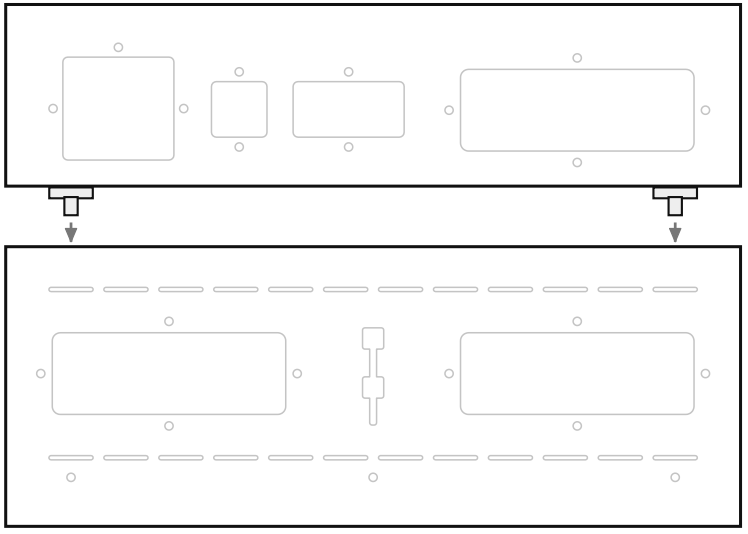
☑ **Lid seats without bowing the walls · no wire pinched**

3.3 Stack

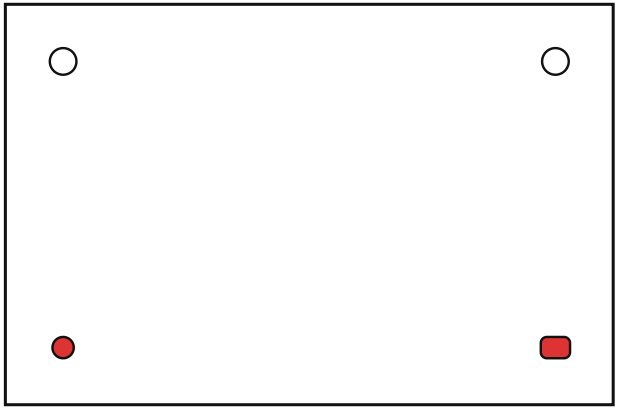


1. Push the 4 printed feet into the control box's base holes.
2. Lower the control box onto the power lid: the front-left peg finds the round hole, the front-right the slot.
3. Banana leads, post to post, colour-matched: 24 V, G1, 12 V, 6 V, G2.

⚠ P14 not modelled yet



front view: control box onto the power lid



power lid top: hole, slot, 2 clearance

☑ **Control box sits square, no rocking**

3.4 Laptop board

board

1x

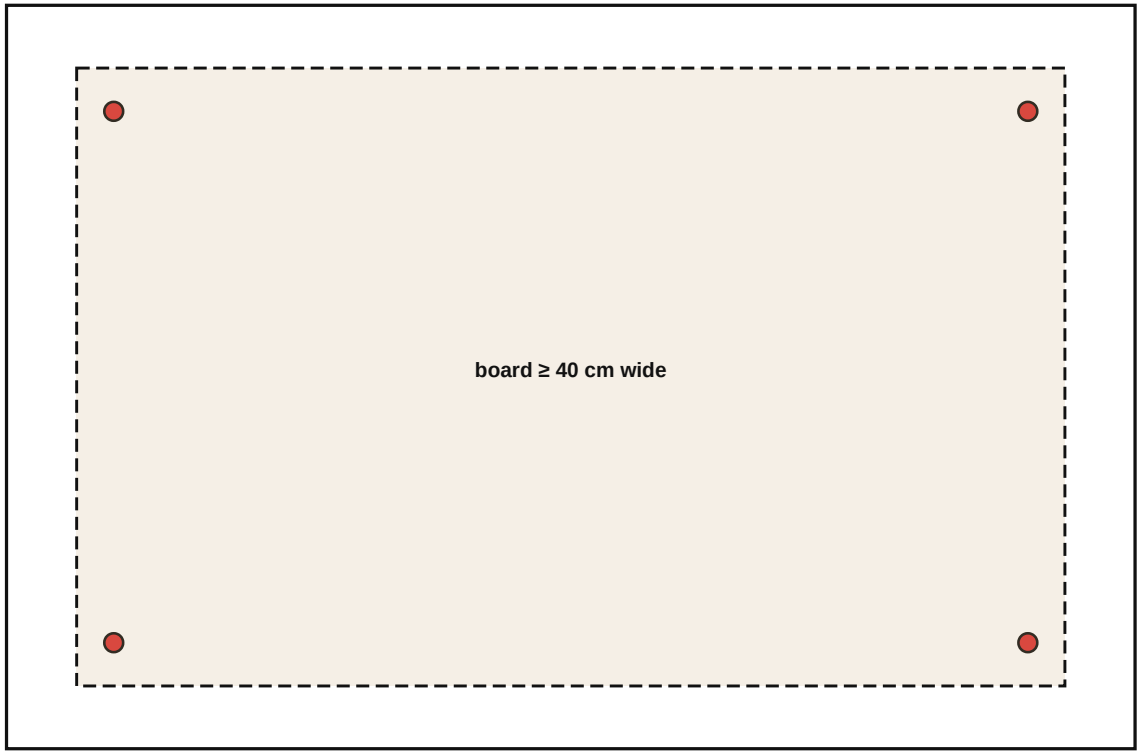
board ≥ 40 cm

B54

have

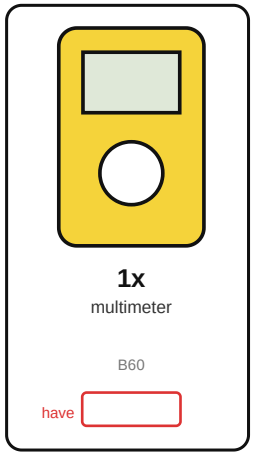
1. A board at least 40 cm wide: on a 38 cm board the lid's corner holes sit only 5 mm in from its edges.
2. Or pads under it. M3 rivnuts in the 4 corner holes if you want it fixed.

Later: not on the path to the first A3



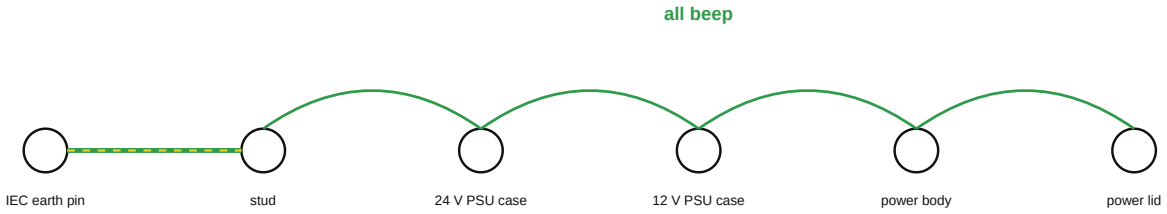
FRONT

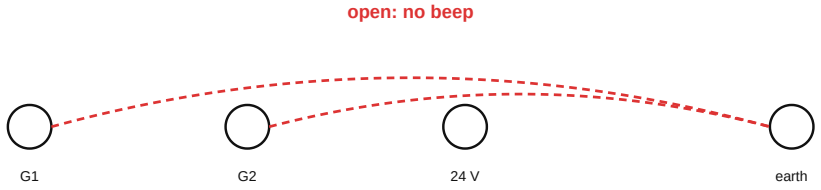
4.1 Checks before motors



1. Unplugged. Earth: IEC earth pin → stud, both PSU cases, power body, both lids, control body. All beep.
2. Isolation: G1 and G2 to earth: open. 24 V to G1: not a short.
3. Power box alone, banana leads out, mains on: every jack at its voltage.

Lids beep only if the bond leads go in (Open questions 5)





☑ **All three pass**

4.2 First power-up

1. Up: USB first, then the toggles. Down: toggles off, then USB.
2. Motors one at a time. If X1 and X2 fight, swap one coil pair on one motor.
3. Endstops before the first homing.

Never plug or unplug a motor with 24 V on

1 USB in
board boots

→ **2 banana leads**

→ **3 24 V toggle**
TB LEDs on

→ **4 6 V toggle**
servo unplugged: 6.0 V on pins 1–2

5 one motor
toggles off to plug; jog 10 mm

→ **6 endstops**
press each by hand: status shows it

→ **7 servo**
pen up/down × 5, no reset

→ **8 10 min run**
drivers warm, fans spin

☑ **Then Stage 5: the first A3**

BOM 1

1 Master copy: the Etch Build Book doc. Status here is as of 2026-10-22.

ID	Item	Spec	Qty	Step	Status
B01	IEC C14 3-in-1 module	inlet + 5x20 fuse + lit double-pole rocker	1	1.4	Check
B02	Fuse 5x20	T3.15 A slow-blow	3	1.4	Need
B03	Latching stop button	LA38-11ZS, Ø22, NC	1	1.4	Check
B04	Panel LED	12 mm metal, 12-24 V, green	2	1.4	Check
B05	Mains wire	H05V-K 1.0 mm²: brown, blue, green/yellow	2 m each	1.1, 1.5	Need
B06	Lever splicer	genuine Wago 221-413	4	1.5	Need
B07	Female spade, insulated	match the IEC tabs (4.8 or 6.3)	10	1.5	Need
B08	Ferrule kit	0.25-2.5 mm², insulated + bare	1	1.5, 1.7, 2.8	Check
B09	Earth stud set	M6 x 30 hex bolt (A2), 3 nuts, 2 serrated washers	1	1.1	Need
B10	Ring terminal M6	insulated, 0.5-1.5 mm²	6	1.1	Need
B11	Terminal shrouds	or heat shrink for IEC tabs, PSU inputs	1 set	1.5	Need
B13	24 V PSU	Mean Well LRS-350-24	1	1.3	Have
B14	12 V PSU	model: read the label	1	1.3	Have
B15	Buck 12 -> 6 V	type: check (XL4015 preferred)	1	1.3	Have
B16	PCT 2x6 splitter	2 x (1 in / 3 out)	1	1.7	Have
B17	DC wire, heavy	1.5 mm² (not 16 AWG), red + black	3 m each	1.7, 2.8	Need
B18	DC wire, light	0.75 mm² silicone, red + black	3 m each	1.7, 2.8	Need black (have red
B19	Ring terminal M4	insulated: blue x10, red x20	30	1.9, 2.6	Need
B20	Fan	80 x 80 x 25, 12 V	2	1.8, 2.10	Have
B21	Fan screw set	M4 x 35 + nyloc + washer	8	1.8, 2.10	Need
B22	M4 nut	on the standoff studs (plain or nyloc: ask)	12	1.2, 2.1	Need
B23	Banana panel screw set	M3 x 10 + washer + nyloc	8	1.9, 2.6	Need
B24	Rubber feet	stick-on, power box	4	0.2	Have
B25	Base plates	440 x 285, 2 mm mild steel (cut)	2	1.2, 2.1	Have

BOM 2

1 Master copy: the Etch Build Book doc. Status here is as of 2026-10-22.

ID	Item	Spec	Qty	Step	Status
B26	M4 standoff	8 mm, male-female	12	1.2, 2.1	Need
B27	M4 screw	M4 x 6 pan head, Phillips (floor into standoff)	12	1.2, 2.1	Need
B28	M4 rivnut	for the Ø6 grid (hole unmeasured)	20	1.3, 2.2	Need
B29	M3 rivnut	for Ø5 holes; 39 needed	50	0.2	Have
B30	Solid rivet	Ø3.2; 8 needed	15	0.3	Have
B31	PSU mount screws	thread + length from datasheets	8	1.3	Need
B32	Stepper driver	TB6600	3	2.3	Have
B33	Controller	FluidNC board	1	2.3	Have
B34	DIN rail	TS35, 8 in (203 mm)	2	2.2	Have
B35	DIN clip	25 x 69, 2 x Ø3.4 at 58.5	5	2.3	Ordered
B36	Clip screw	M3 flat head x 8 (unverified)	12	2.3	Need
B37	Heat-set insert	M3 x 4, OD 4.3	20	2.3	Have
B38	Board screw	M3 x 6 pan head	4	2.3	Need
B39	TB6600 flange screw	M3 x 10 + nut	6	2.3	Need
B40	DIN rail screws	M4 x 10 + washer, + 2 nyloc	4	2.2	Need
B41	GX16 socket + plug	3 x GX16-5, 1 x GX16-4, 2 x GX16-3	6	2.5	Check
B42	Panel screw	M3 x 6 countersunk	40	1.4, 2.5, 3.1	Need
B43	USB-C pass-through	part: decide	1	2.7	Decide
B45	Signal wire	0.25-0.5 mm², several colours	1 set	2.9	Check
B46	Stepper cable	4-core screened	10 m	2.9	Check
B47	Endstop cable	2-pair screened	4 m	2.9	Check
B48	Servo cable	3-core 0.5 mm² silicone, high-flex	2 m	2.9	Check
B49	Capacitor	2200 µF 16 V low-ESR, at the servo	2	2.8	Check
B50	Cable management	ties, mounts, sleeving, heat shrink	1 set	2.9	Check

BOM 3

1 Master copy: the Etch Build Book doc. Status here is as of 2026-10-22.

ID	Item	Spec	Qty	Step	Status
B51	Lid screw	M3 button head (length unverified)	12	3.2	Need
B53	Banana lead	4 mm, colour-matched	5	3.3	Check
B54	Laptop board	>= 40 cm wide	1	3.4	Later
B60	Multimeter	continuity + DC 0.01 V	1	0.1	Have
B61	Rivnut tool	M3 + M4	1	0.1	Have
B62	Ferrule crimper	0.25-6 mm²	1	0.1	Have
B63	Terminal crimper	insulated rings + spades	1	0.1	Have
B64	Labels	wire markers or label maker	1	0.1	Have

Prints 1

1 Master copy: the Etch Build Book doc. Status here is as of 2026-10-22.

ID	Part	Script → STL	Qty	Step	Status
P01	IEC panel	not modelled; needs P02 results	1	1.4	To model
P02	Fit coupons C + D	fit_kit.py -> fit_kit/	1 set	1.4	Ready
P03	PSU brackets	not modelled; needs PSU hole patterns	2 sets	1.3	To model
P04	Base plate rest pads Ø20 x 5	not modelled; Al plate only	6	1.2, 2.1	To model
P05	PSU panel	link_panel.py (psu)	1	1.9	Printed
P06	Receiving panel	link_panel.py (ctrl)	1	2.6	Printed
P07	TB6600 plate, 89 pitch	din_mounts.py -> tb6600_plate.stl	3	2.3	Ready
P08	Board plate	din_mounts.py -> board_plate.stl	1	2.3	Ready
P09	GX plates	gx_plates.py -> gx_plate_*.stl	3	2.5	Ready
P10	USB / antenna panel	not modelled; gate B43	1	2.7	To model
P11	Bend-jig sleeves Ø16/24/40	bend_jig.py -> bend_sleeve_d*.stl	4/4/2	2.9	Ready
P12	Meccano window panels	window_meccano.py -> meccano_*.stl	3	3.1	Ready
P13	Lid washers	not modelled	14	3.2	To model
P14	Stacking feet	not modelled	4	3.3	To model
P15	Wire guides	wire_jig.py	as needed	2.9	Printed
P16	Logo spine + plate	window_inserts.py (logo)	1	later	Later